



CHAPTER 13: Production Planning

Planning your production 507

Production Planning is the process of modeling available capacity to balance workloads while effectively meeting demand. With a strong production plan you can minimize the time and costs of production and best utilize your resources.

In Maestro, our Production Planning application helps you create feasible production plans by using constraints to model capacity on production wheels. See ["Assessing changeover" on page 538](#).

Built as a complement to the Capacity Planning (Constraints) application, Production Planning enables you to:

- Use production wheels to reduce costs and increase your operational efficiency. See ["Production wheels" on page 514](#).
- Use production recurrence to align production frequency to a product segmentation strategy. See ["Production recurrence" on page 533](#) and ["Review and update product segmentation" on page 508](#).
- Use product grouping to reduce setup and clean up activities in production. See ["Production groups" on page 543](#).
- Use product sequences to predefine the order of manufacture to further improve your capacity utilization. See ["Part sequencing" on page 526](#).

There are multiple resources available in the Production Planning application. For a summary of what you can do in each resource, see ["Production Planning resources" on page 555](#).

Planning your production

Use the following steps to define and manage production in your system. Click a step to jump to that section.

Review and update your product segmentation

If your company uses segmentation strategies in production, start by reviewing the segmentation applied to parts produced using production wheels.

Define parameters for production

Define the parameters for production wheels, production groups, production recurrences, and segments assigned to parts produced using production wheels.

Review constraint usage for all production wheels in your system

Identify possible planning issues by reviewing constraint consumption by part or wheel, comparing constraint allocation, and checking the sequencing of parts allocated to a production wheel.

Adjust production parameters as needed

If needed, adjust production parameters to optimize your plan.

Planning for different supply order types

Production planning supports both planned orders and the following types of scheduled receipts (SRs):

Scheduled SRs: scheduled receipts that can exceed constraint capacity in a production wheel. Scheduled SRs are placed in a production wheel based on their start date and are produced until the order is complete or the due date is reached.

Non-reschedulable SRs: scheduled receipts that respect the allocated constraint capacity of a production wheel. In a production cycle, they consume constraint after scheduled SRs but before planned orders and reschedulable SRs. If a current production cycle doesn't have sufficient constraint, non-reschedulable SRs are deferred to the next recurring production cycle with available constraint.

Reschedulable SRs: scheduled receipts can be moved within production cycles, deferred to future cycles, or pulled in from future cycles if sufficient constraint is available. When a reschedulable SR is pulled in from a future cycle, the expedite limit date (SupplyStatus.ExpediteLimit) is extended to the start of that cycle to prevent supplies from being moved too early.

Caution: Running production cycles that include both scheduled SRs and non-reschedulable SRs might cause incorrect changeover calculations. It is recommended you avoid having these two SR types overlap in the same cycle.

Sorting SRs in a production wheel

SRs are placed into production cycles based on what type of SR they are:

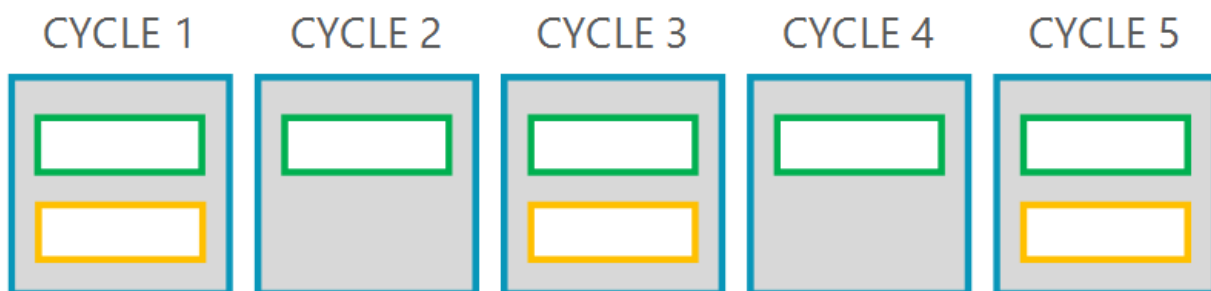
- Scheduled SRs always take precedence in a cycle and are planned for first.
- Non-reschedulable SRs are produced before planned orders and are automatically placed at the start of a cycle.
- Reschedulable SRs and planned orders are loaded into a cycle based on assigned part sequence.

Between supply orders of the same type, part sequencing, wheel and group priorities are used to sort the orders.

Review and update product segmentation

Segmentation strategies can help you organize the production frequency you want to apply to parts made in a production wheel. The segments use colors to group parts by importance, enabling you to tag a set of parts with the same production recurrence. See ["Production recurrence" on page 533](#).

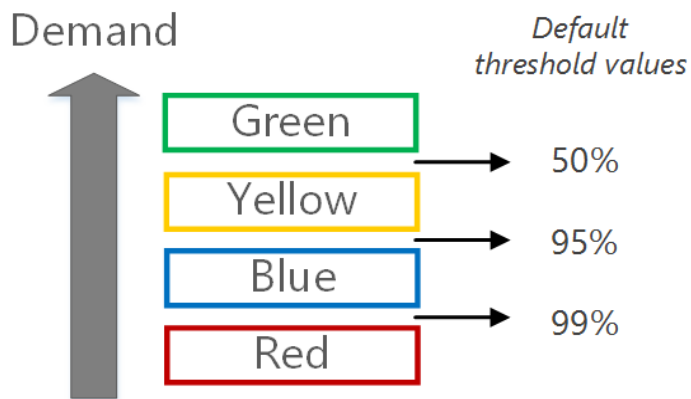
For example, the diagram below shows green parts and yellow parts produced over 5 cycles in a production wheel. The production recurrence for green parts is set to every cycle and as every other cycle for yellow parts.



By default, the segmentation strategy in the Production Planning application has four segments. Each segment is assigned a threshold value that represents a range of cumulative demand for that part.

The diagram below shows the default segments ordered by cumulative demand with the green segment representing the highest levels of demand. Default threshold values are set between each segment. For example, parts with a cumulative demand between 50% and 95% are tagged

as yellow parts. The default threshold values can be changed. See ["Changing or overriding segment thresholds"](#) on page 511.



Use the following steps to review and update the segmentation applied to your production:

Step	Description
1. Sort by Units or Revenue	Depending on how you want to view the demand for a part and the segment assigned to that part, you can sort demand by units or by revenue. See "Sort demand data" on page 510.
2. Optionally, change the production recurrence for a segment	Segments are assigned a production recurrence that applies to all the parts tagged with that segment. To change how often parts in a segment are produced, you can assign a different production recurrence to the segment. See "Assign a production recurrence" on page 536.
2. Adjust thresholds	Segments are assigned to parts based on what threshold level the demand percentage for that part meets. By default, the following thresholds are assigned: • Green-Yellow threshold: 50% • Yellow-Blue threshold: 95% • Blue-Red threshold: 99% See "Change a segment threshold" on page 511.
3. Accept new segments	After making adjustments to any of the thresholds, you must accept the changes before the new segments are assigned to parts for production. See "Accept segment threshold changes" on page 512.
4. Override a segment	Optionally, you can override individual segment assignments and change which segment a part is assigned. See "Override a segment assignment" on page 513.

Viewing segmentation data

When you review the segmentation applied to parts in your system, you can drill to more details about the production of each part, switch how demand values are sorted, or change the time periods that the demand data displays for.

Show part details

You can review details for the part sources associated with a segment including the production wheel, group, and recurrence associated with each part.

1. In the **Production Planning Parameters** workbook, click the **Segments** worksheet tab.
2. Click a linked value in the **Part Sources - Per Segment** column.

Sort demand data

You can sort cumulative demand data by units or by revenue to assess how segmentation is applied. By default, the data displays in units.

► Sort demand data by revenue

To sort by revenue, show the demand revenue in the worksheet.

- In the **Production Planning Segmentation** workbook, do one of the following:
 - On the worksheet toolbar click **Sort by:** and select **Revenue**.
 - In the **Data Settings** dialog box, select **Revenue** in the **Sort by** field.

► Sort demand data by units

To sort by units, hide demand revenue in the worksheet.

- In the **Production Planning Segmentation** workbook, do one of the following:
 - On the worksheet toolbar click **Sort by:** and select **Units**.
 - In the **Data Settings** dialog box, select **Units** in the **Sort by** field.

Define when demand data displays for

You can also define the periods that demand data shows for by selecting which calendar to use and the number of future and previous calendar periods that display in the worksheet.

► Define the periods to show demand data for

1. In the **Production Planning Segmentation** workbook, on the workbook toolbar, click **Data Settings** .
2. Type a value for **Future Periods**.

3. Type a value for **Previous Periods**.
4. In the **Calendar** field, select a value from the drop-down list.

Changing or overriding segment thresholds

The Maestro Production Planning application is configured with default thresholds between segments. The table below identifies the default threshold limits:

Threshold	Cumulative Demand %
Green-Yellow	50%
Yellow-Blue	95%
Blue-Red	99%

If a part spans two segment thresholds, Maestro assigns the part to the higher segment.

You can customize how segments are assigned to parts by changing the threshold limits or by overriding the segment assigned to a specific part. For more on how segmentation affects production, see ["Review and update product segmentation" on page 508](#).

Change a segment threshold

To change the threshold value for a segment, you can specify a new value in the Data Settings dialog box or use the segment arrows in the Production Planning Segmentation workbook.

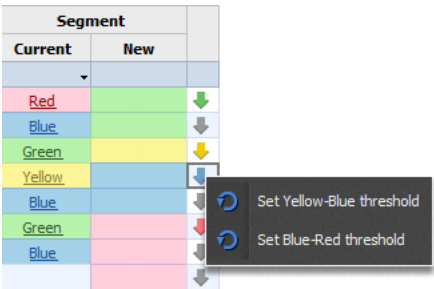
► Change a threshold using the Data Settings

1. In the **Production Planning Segmentation** workbook, on the workbook toolbar, click **Data Settings** .
2. Type a value for any of the following threshold fields:
 - **Green-Yellow Threshold(%)**: This defines what parts are assigned to the green segment. Values above this number are assigned to the yellow segment.
 - **Yellow-Blue Threshold(%)**: This defines what parts are assigned to the yellow segment. Values above this number are assigned to the blue segment.
 - **Blue-Red Threshold(%)**: This defines what parts are assigned to the blue segment. Values above this number are assigned to the red segment.
3. Click **OK**.

► Change a threshold using segment arrows

Segment arrows display for each part assigned to a segment. You can set the value of the cumulative demand for a part as a threshold value using the colored segment arrows.

When you click the colored segment arrow in a row, the value of the cumulative demand of the part in that row is assigned as the new threshold value. For arrows at the boundary between two thresholds, you can select which threshold you want to set.



1. In the **Production Planning Segmentation** workbook, select the **Part** with the **Cumulative Demand** you want to set as a threshold value.
2. Do one of the following:
 - In that row, click the segment arrow .
 - In a row that defines the boundary between two thresholds, click the segment threshold you want to set in the list.



Accept segment threshold changes

Once you are satisfied with the changes to the threshold levels, commit those changes by running the command to accept the new segment assignments. The current segment column is updated with the new threshold values.

Cumulative	Segment		
	Current	New	
24.04%	Red	Green	↓
42.14%	Blue	Green	↓
54.72%	Green	Yellow	↓
66.27%	Yellow	Blue	↓
76.17%	Blue	Blue	↓
85.84%	Green	Blue	↓
94.04%	Blue	Blue	↓
100.00%		Red	↓

Accept new segments →

Cumulative	Segment		
	Current	New	
24.04%	Green	Green	↓
42.14%	Green	Green	↓
54.72%	Yellow	Yellow	↓
66.27%	Blue	Blue	↓
76.17%	Blue	Blue	↓
85.84%	Blue	Blue	↓
94.04%	Blue	Blue	↓
100.00%	Red	Red	↓

1. In the **Production Planning Segmentation** workbook, on the workbook toolbar, click **Run Command** , and then click **Accept New Segments**.
2. In the **Run Command** dialog box, click **Yes**.

► Reset threshold levels

At any time you can reset threshold levels to the default values specified by the workbook author. The thresholds can return to their defaults even after you have run the command to accept new segment values.

- In the **Production Planning Segmentation** workbook, click **Reset Thresholds** on the worksheet toolbar.

Override a segment assignment

Segments are used to control when a set of parts is produced in a production wheel. You can change which segment a part is assigned to by overriding its segment assignment.

1. In the **Production Planning Segmentation** workbook, find the part you want to override.
2. Click the value in the **Segment - Current** column.
3. Click **Value** in the **Override Segment** worksheet.
4. Select a new segment in the list.
5. Save your changes.



Note: If you have two segments with the same name at different sites and you have accidentally selected the segment at the wrong site, select the correct site in the **Segment - Site** column.

Define and adjust parameters for production

When you define the parameters for a production plan, you start with setting up the production wheel or wheels the plan uses, assign production recurrences to segments (or directly to parts), review and allocate constraint for changeovers, and assign parts to production groups.

Use the following steps to define the parameters to use for production:

Step	Description
1. Create or update your production wheels	Production wheels are used to organize and model your production. Create a new wheel or modify an existing wheel to meet your production needs. See "Production wheels" on page 514 .
2. Review production recurrences	Production recurrences determine the production frequency of a part. The recurrence might be defined for a part or it might be defined for the segment assigned to a part. See "Production recurrence" on page 533 .
3. Assess changeover amounts and group priorities	Assess the amounts of constraint allocated to cycle setup and cleanup and to changeovers between parts. See "Assessing changeover" on page 538. Review production priorities for parts in a production group. See "Production priorities" on page 542. If you have preplanning enabled in your system, production groups are formed to optimize capacity. See "Pre-planning and production wheels" on page 540.
4. Create or update the production groups	Optionally, you can organize parts into production groups to use in production wheels. See "Production groups" on page 543. Production groups are an additional way to organize part production that is focused on minimizing changeover costs and increasing efficiency.

Production wheels

Production wheels organize the production of multiple parts using the same equipment in repeating cycles.

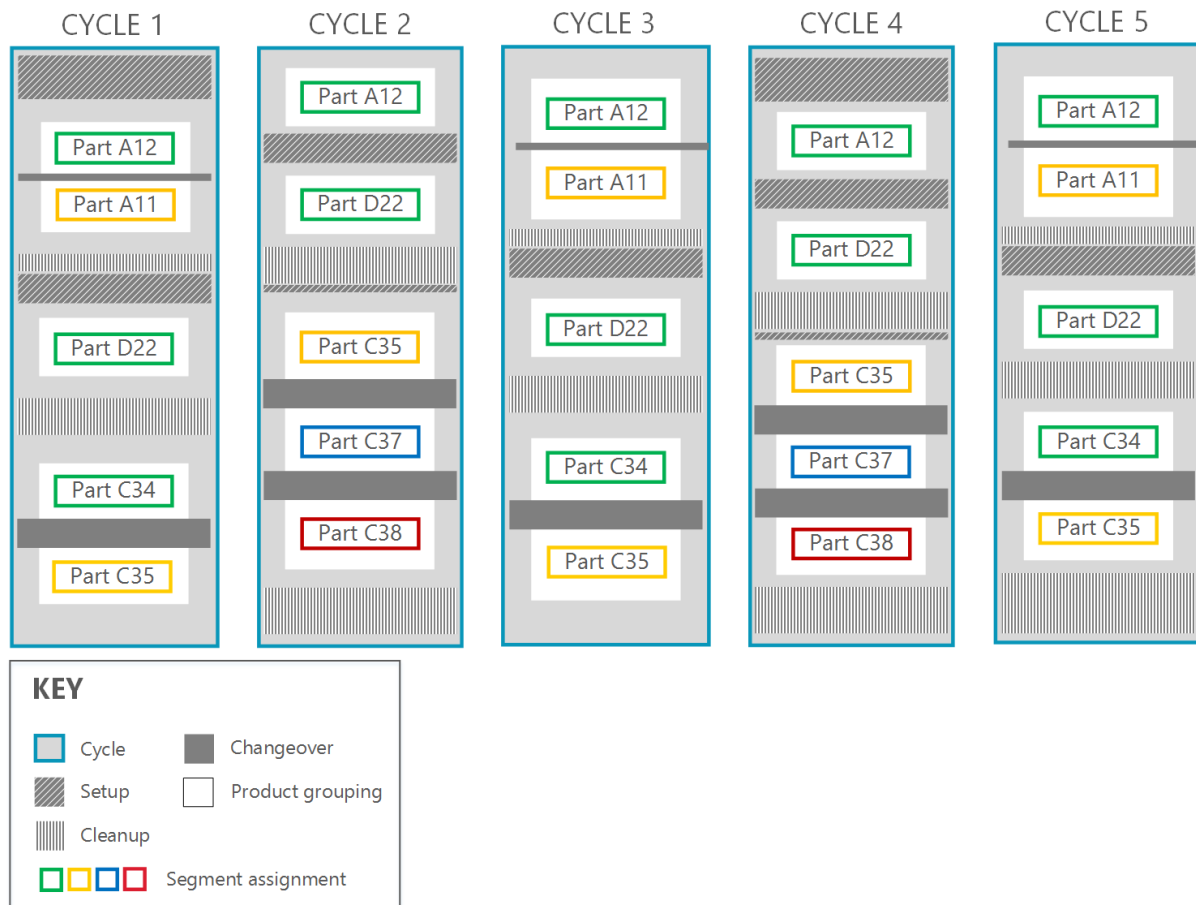
Use production wheels to help you:

- Reduce changeover costs
- Reduce waste
- Level inventory levels
- Increase operational stability and production flexibility

In the Production Planning application, production wheels use constraint to model the production of parts including downtime required for activities such as setup or cleanup.

Each production wheel is divided into cycles. A fixed number of cycles run for the duration of a production wheel. See ["Defining production wheel and cycle durations" on page 518](#).

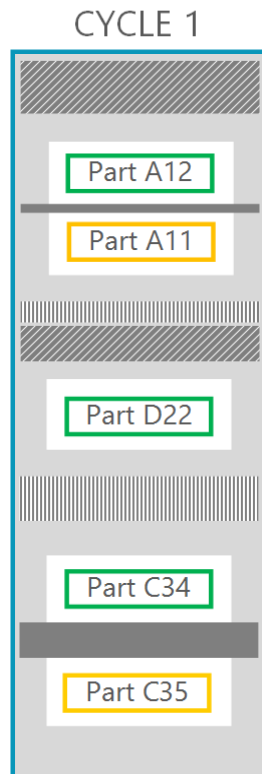
The diagram below shows a production wheel with 5 cycles. Parts in each cycle are grouped (See [Production groups](#)) and the constraint required for changeovers, setup, and clean up is identified. Parts in this wheel are also tagged by segment (See [Segmentation](#)).



Note: Production wheels uses constraint consumption dates, such as **ConstraintUsed.Date**, to determine when production is on time.

In the diagram below constraint is allocated to set ups before and after producing a group, clean ups after producing groups, and to the changeovers between the production of parts within a group.

For more on constraints in a production wheel, see ["Managing supply and constraints in a production wheel" on page 522](#).



Use the following steps to create or update a production wheel:

Step	Description
1. Create or modify a production wheel	Create a new production wheel or modify an existing production wheel and specify the wheel duration and cycle lengths. See "Defining production wheels" on page 517 .
2. Assign a constraint to the wheel	Assign a constraint to a production wheel. This constraint is divided among the cycles in the wheel. See "Assessing changeover" on page 538 .
3. Associate part sources with the wheel	Parts assigned to be produced in the wheel automatically associate part sources with the wheel. If you are using segmentation strategies, you can manually associate part sources with a segment and then assign that segment to the wheel. See "Associate part sources with a production wheel" on page 525 .
4. Optionally, generate part sequences to include sequencing	Control the frequency of production of a part by assigning sequence numbers to parts. See "Part sequencing" on page 526 .

Defining production wheels

When you create or modify a production wheel, in addition to defining basic properties for the wheel such as its name and site, you must also define how long the wheel runs for, its priority in a production sequence, and how the wheel reacts to unsatisfied supply or unused constraint.

If you want to see more details about either a production wheel or the total constraint assigned to that production wheel, you can drill to more detailed information.

For wheels that are no longer needed, you can delete the wheel from your system.

► Create or modify a production wheel

1. In the **Production Planning Parameters** workbook, click the **Wheels** worksheet tab.
2. To create a new wheel, on the toolbar, click **Insert Record** .
3. To modify a wheel, select the record for that wheel.
4. In the **Insert Record - Production Wheel** form, specify the following:
 - **Name:** The name of the production wheel. If you are typing a new name, you cannot use the same name as any other production wheel in your system that is at the same site.
 - **Site:** The site for the production wheel.
 - **Type:** The control table rule you want applied to unsatisfied supply quantities or to unused constraint in the production cycle. See ["Managing supply and constraints in a production wheel" on page 522](#).
 - **Description:** A description of the production wheel.
 - **Priority:** The number value for the priority. This priority determines what order the production wheel is processed when there are multiple production wheels running in the same time period.

Cycle

- **Calendar:** The calendar used to determine the production wheel duration.
- **Interval:** The number of calendar intervals you want in the production cycle. When combined with the calendar this determines the size of a cycle in the production wheel. For more information, see ["Defining production wheel and cycle durations" on page 518](#).

Dates

- **Effective In Date:** The date you want the production wheel to start.
 - **Effective Out Date:** The date you want the production wheel to finish.
5. Click **OK**.
 6. Save your changes.

► See more about a production wheel

You can drill to more detailed information about a production wheel.

1. In the **Production Planning Parameters** workbook, click the **Wheels** worksheet tab.
2. Click a value in the **Wheel - Name** column.
The **Production Planning Manager** workbook opens. See "[Production Planning Manager workbook](#)" on page 555.

► See more about constraints

You can drill to more detailed information about the constraint assigned to a production wheel.

1. In the **Production Planning Parameters** workbook, click the **Wheels** worksheet tab.
2. Click a value in the **Associated Constraint** column and select one of the following:
 - **Constraint Properties:** See detailed data about constraints and how they are configured.
 - **Constraint Manager:** See a summary of constraints and the scheduled usage for those constraints. You can drill to different levels of detail about constraints from this workbook.

► Delete a production wheel

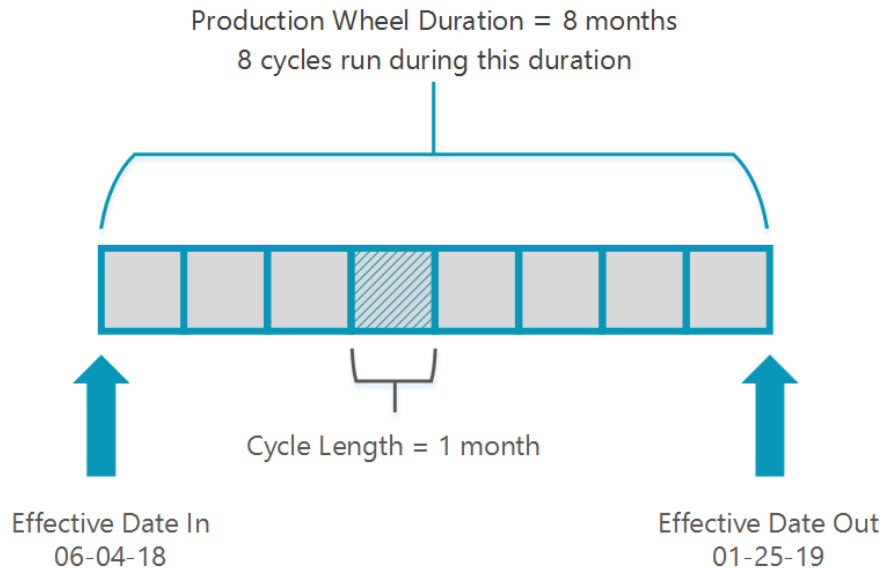
If a production wheel is no longer required or updating it is not feasible, then you can delete the production wheel record from your Maestro system.

1. In the **Production Planning Parameters** workbook, click the **Wheels** worksheet tab
2. Select the row for the production wheel you want to delete.
3. On the toolbar, click **Delete Record** .
4. Save your changes.

Defining production wheel and cycle durations

You can customize the production frequency of parts by defining the duration for a production wheel and the number of cycles that run in the wheel. The duration of a production wheel is determined by its effective in and out dates. Based on the cycle length you specify, Maestro determines the number of cycles that run in the wheel.

For example, in a production wheel that spans 8 months and has a cycle length of 1 month, Maestro schedules 8 cycles to run in that wheel.



The production cycles should be aligned with the effective dates and calendar assigned to the applicable constraints. See ["Constraint Properties" on page 363](#).

For some wheels, Maestro shifts the effective in and out dates to fit the cycle length that you specify. See ["Calibrating production wheel durations" on page 520](#).

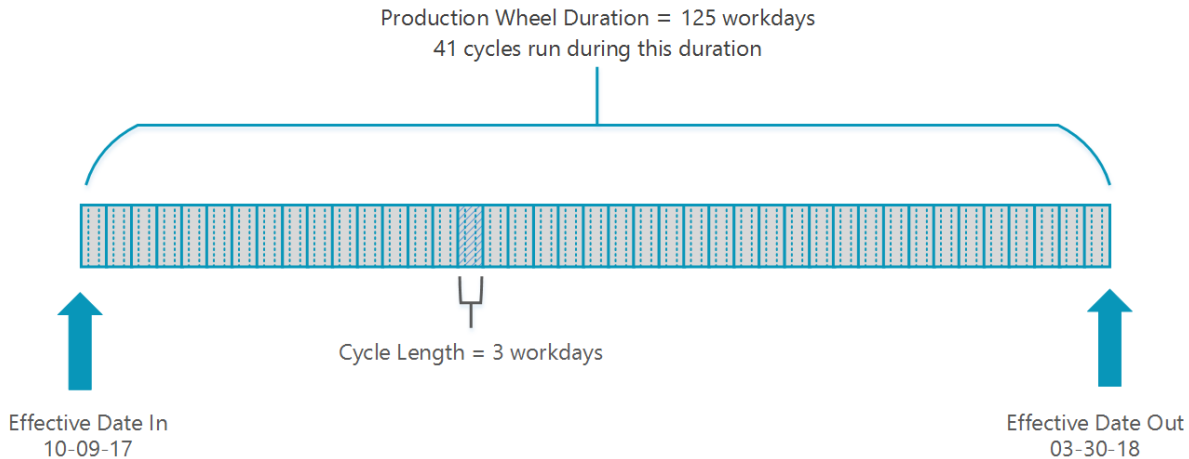
► Define the duration for a production wheel

1. In the **Production Planning Parameters** workbook, select the **Wheels** worksheet.
2. In the worksheet select the production wheel you want to define the duration for.
3. Specify a date to start the production wheel on in the **Effective Date - In** column.
4. Specify a date you want to stop the production wheel on in the **Effective Date - Out** column.
5. Save your changes.

Cycle lengths

The cycle length is set by the number of intervals in a cycle and the assigned cycle calendar.

For example, if you set the cycle interval to 3 and the calendar to Workday, then the cycle length is 3 weekdays. For a production wheel that runs for 125 workdays, 41 cycles are run over that duration.



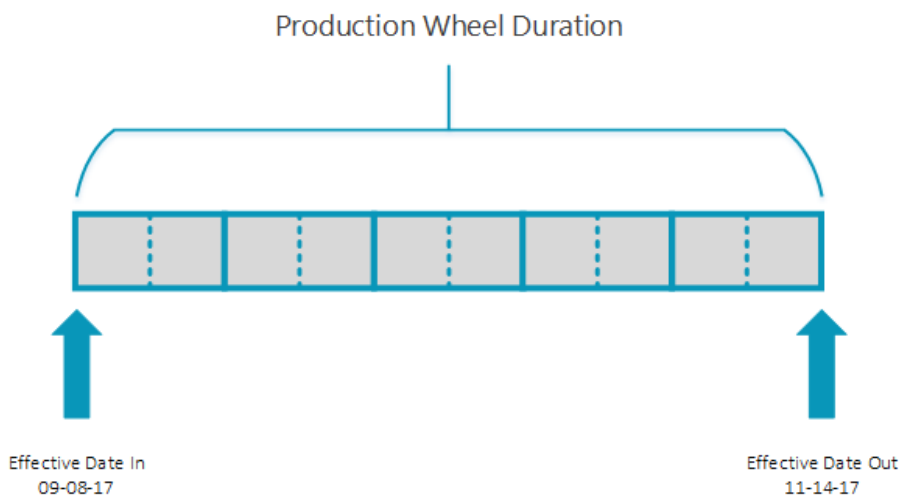
► Define the length of a cycle

1. In the **Production Planning Parameters** workbook, click the **Wheels** worksheet.
2. In the worksheet, select the production wheel you to define the cycle length for.
3. Type a value in the **Cycle - Interval** column.
4. Select a calendar in the **Cycle - Calendar** column.
5. Save your changes.

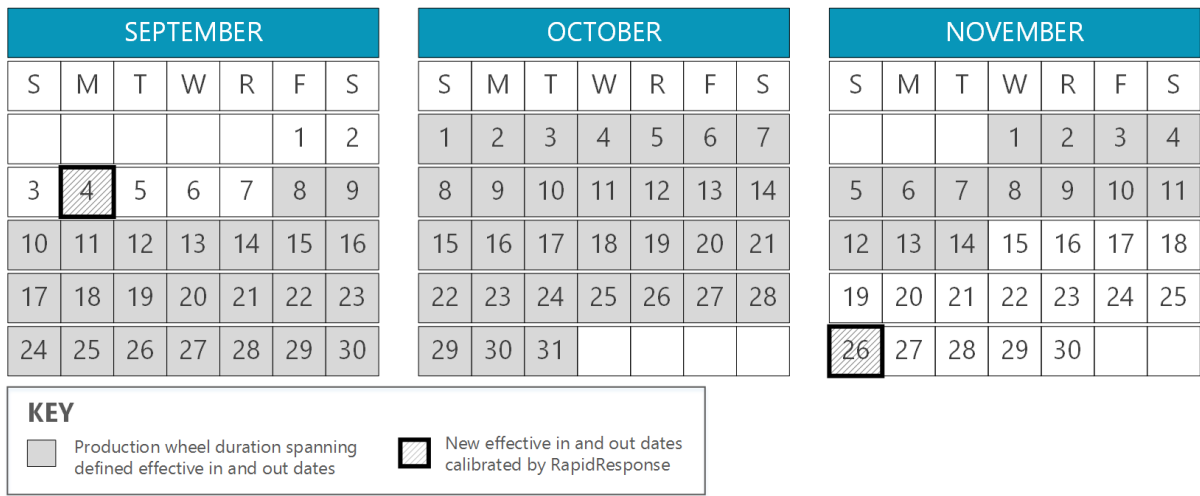
Calibrating production wheel durations

The duration for a production wheel is determined by its effective in and out dates. However, based on the cycle length, Maestro might shift effective dates to avoid having any partial cycles in the wheel and to effectively optimize production in the wheel.

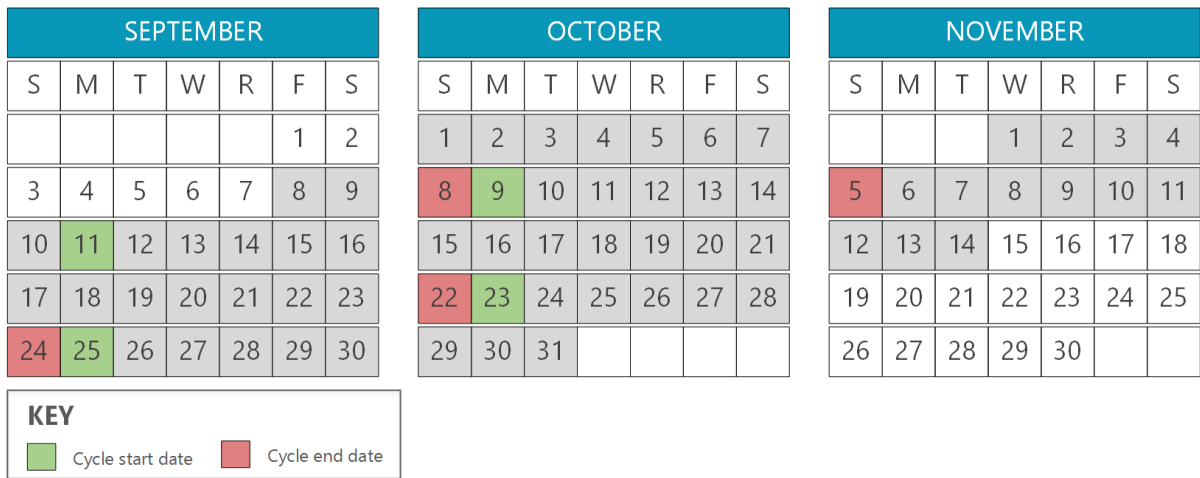
For example, the production wheel below runs from September 8 to November 14. Using a Week calendar, the wheel duration is about 7 weeks.



When a cycle length of 2 weeks is assigned to the wheel, Maestro shifts the effective in and out dates to September 4 and November 26.



If the effective dates are not shifted, only four cycles can run in the wheel. The first cycle can't start until September 11 and the last cycle finishes on November 5. As the cycle duration is 2 weeks and partial cycles can't run, the production potential of the week of November 6 is lost.



Instead, Maestro shifts those dates enabling 6 cycles to run for that wheel duration.

SEPTEMBER							OCTOBER							NOVEMBER						
S	M	T	W	R	F	S	S	M	T	W	R	F	S	S	M	T	W	R	F	S
					1	2	1	2	3	4	5	6	7				1	2	3	4
3	4	5	6	7	8	9	8	9	10	11	12	13	14	5	6	7	8	9	10	11
10	11	12	13	14	15	16	15	16	17	18	19	20	21	12	13	14	15	16	17	18
17	18	19	20	21	22	23	22	23	24	25	26	27	28	19	20	21	22	23	24	25
24	25	26	27	28	29	30	29	30	31					26	27	28	29	30		

KEY
 Cycle start date
 Cycle end date

Managing supply and constraints in a production wheel

When you create or modify a production wheel, you determine which constraint is assigned to the wheel. One of the parameters you must also specify is how the wheel handles unsatisfied supply quantities or unused constraint amounts.

Unsatisfied supply

Defer: Unsatisfied supply quantities are handled in the next cycle where the part can be built.

Ignore: Unsatisfied supply quantities are ignored and not built in the cycle. Instead, those supplies are assigned an available date of 'Future'

Unused constraint

Satisfy: Unused constraint is used to satisfy supplies that are normally produced in future cycles that are eligible to be built for the current cycle of the wheel.

Ignore: If the constraint used in the current cycle doesn't meet the minimum constraint consumption value, then the minimum value is ignored.



Note: Scheduled SRs and non-reschedulable SRs are not affected by the parameters set for unsatisfied supply or unused constraint.

Assign a constraint to a production wheel

You can assign, change, or remove any constraint assigned to a production wheel using the **Assign Constraint** form.

1. In the **Production Planning Parameters** workbook, click the **Wheels** worksheet tab.
2. Select the production wheel you want to modify.
3. In the **Associated Constraint - Assign** column, click **Assign Constraint** .
4. In the **Assign Constraint** form, click the **Constraint** field and select a constraint to assign to the production wheel.
5. In the **Minimum Constraint Consumption** field, type a value to set for the minimum amount of constraint that can be consumed in the production wheel.
6. To remove the constraint assigned to the production wheel, select the **Remove Constraint assignment** checkbox.
7. Click **OK**.

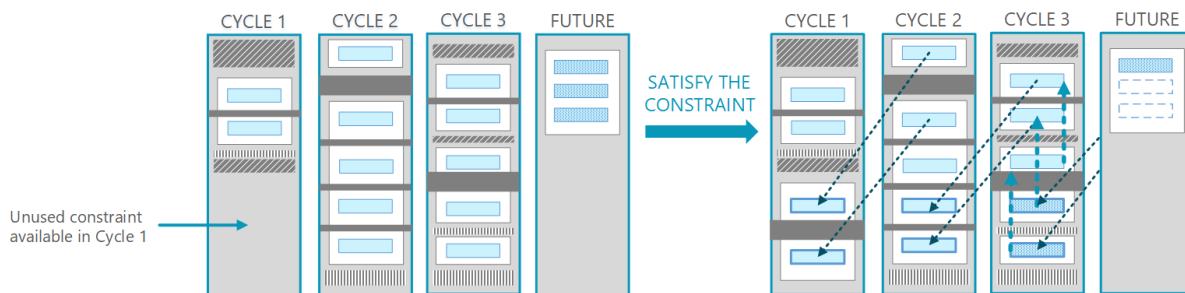
Handle unused constraint amounts

There are two ways to handle unused constraint remaining after all supply quantities have been produced in a production wheel, you can choose to satisfy the constraint or to ignore it.

Satisfy the constraint

If parts produced in future cycles are eligible to build in the current cycle, those parts are assigned to the current cycle.

In the example below, unused constraint in Cycle 1 is satisfied by early supplies from Cycle 2. Cycle 2 now has capacity and early supplies from Cycle 3 are moved into Cycle 2. The newly available capacity in Cycle 3 is satisfied by moving supplies in from the Future.

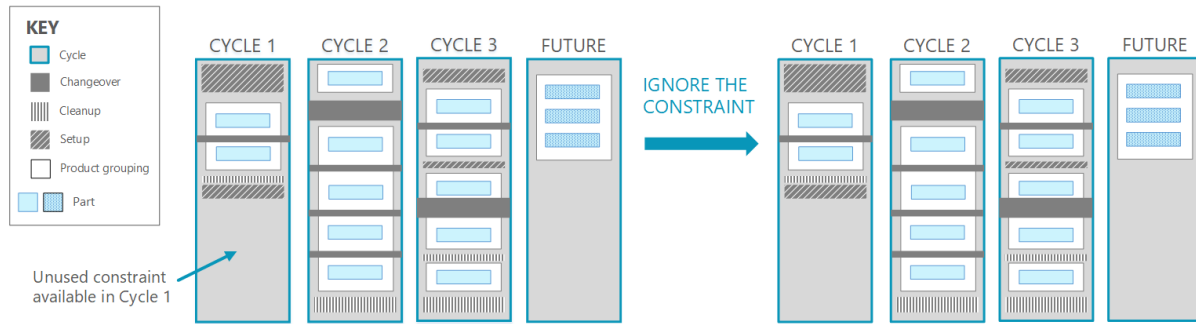


Ignore the constraint

No additional parts are moved from future cycles. The example below shows that the production wheel continues production with unused constraint in Cycle 1.



Note: The minimum constraint consumption limit is ignored in Cycle 1.

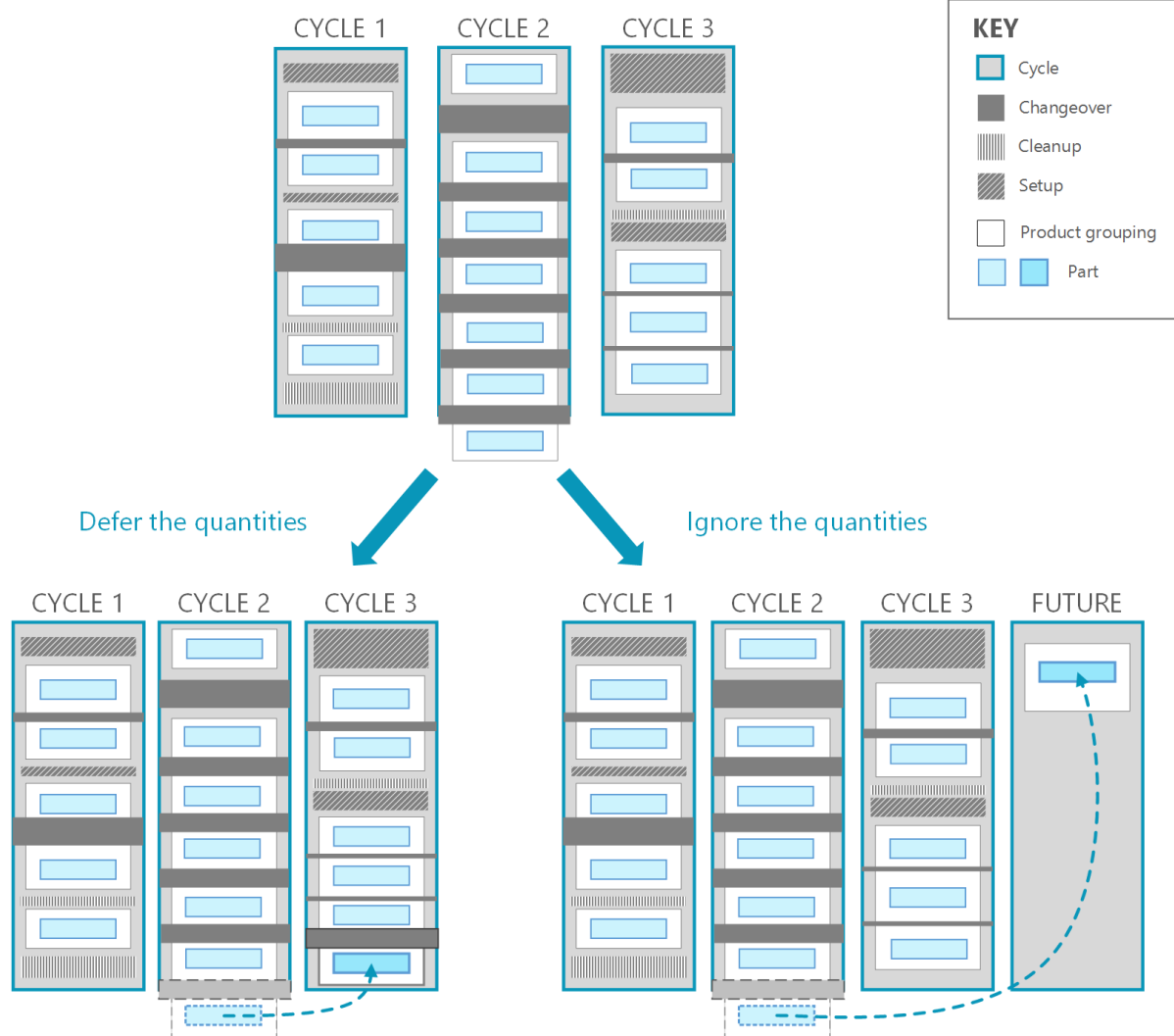


Note: Production groups have a maximum constraint consumption value. See ["Maximum constraint consumption" on page 544](#).

Handle unsatisfied supply quantities

To handle unsatisfied supply quantities assigned to a production cycle that cannot be built because the cycle has run out of constraint, choose from two options:

- **Defer the quantities:** Unsatisfied supply quantities are handled in the next cycle where they can be built. In the diagram below, that is in Cycle 4.
- **Ignore the quantities:** Unsatisfied supply quantities are ignored and not built in any current cycles. Instead that supply is assigned to an available date of 'Future'.



► Manage unused constraint and unsatisfied supply

1. In the **Production Planning Parameters** workbook, click the **Wheels** worksheet tab.
2. Select the production wheel you want to modify.
3. Select a value in the **Type** column.
4. Save your changes.

Associate part sources with a production wheel

Part sources are automatically associated with a production wheel when you assign parts to be produced in that group. To maximize the efficiency of constraint utilization in production planning, include the highest priority part source in the wheel.

1. In the **Production Planning Parameters** workbook, click the **Wheels** worksheet tab.
2. Select the production wheel you want to assign parts to.

3. To associate part sources from the same site as the selected wheel, click the value in the **Part - Sources** column.
Only parts at that site display in the **Part Sources** worksheet.
4. To associate part sources from any site, click the **Add** button  in the **Part - Sources** column.
Parts for all sites display in the **Part Sources** worksheet.
5. For each part you want to produce in this wheel, select the wheel from the list in the **Production Wheel - Name** column.
6. Save your changes.



Note: If you have two production wheels with the same name at different sites and you have accidentally selected the wheel at the wrong site, you can change the site for the wheel using the **Production Wheel - Site** column.

Part sequencing

You have the option of deciding when a specific part is produced in relation to another part by assigning part sequences. For example, a part assigned a sequence number of 1 is planned before a part with a sequence number of 2. Part sequencing takes precedence over production group assignments.

Maestro sorts production using the following criteria:

1. **Production wheel priority:** If assigned, this number determines when the production wheel is processed when there are multiple production wheels assigned to be produced in the same time period.
2. **Production wheel name:** If two or more production wheels are assigned the same priority and are assigned to be produced in the same time period, Maestro uses the wheel name to determine the production order for the wheels. Names are sorted alphabetically.
3. **Supply demand date:** The date that supply is required for production.
4. **Part sequence:** The sequence number assigned to a part.
5. **Production group priority:** The priority assigned to a production group. When multiple production groups are assigned to be produced in the same time period, the priority is used to determine which group runs first.



Caution: Material availability impacts how supply sequencing works in production planning. For example, if material is unavailable at a designated point in the wheel, supplies may not follow the above sort order. The late-available component for a supply may move that supply to a later production cycle which can then impact subsequent cycles. Or components may arrive later in the same cycle, such as on the last day, thereby affecting the constraint usage in that cycle.

► View the part sequences assigned to a production wheel

1. In the **Production Planning Parameters** workbook, click the **Wheels** worksheet tab.
2. Select the row for the wheel you want to view part sequences for.
3. Click the value in the **Part - Sequences** column to open the **Part Sequences** worksheet.
By default, all part sequences are set to a default value of 0. Parts assigned a sequence of 0 are highlighted in yellow in the **Part Sequences** worksheet.

Part sequencing and supply order types

Part sequencing is applied directly to a part, however different types of supply orders can demonstrate modified sequencing behavior when included in production wheels.

Between different supply order types:

- Individual scheduled SRs ignore sequencing and are always put into a cycle based on their start date and continue being produced until they are completed or their due date is reached.
- Individual non-reschedulable SRs follow part sequence but are always built before any planned orders in the cycle.

Between supply orders of the same type:

- Multiple scheduled SRs allocated to the same constraint bucket are prioritized by part sequence.
- Multiple non-reschedulable SRs within the same cycle are sorted against each other by part sequence.

Generating part records

The Part Sequences worksheet displays records for all the parts assigned to a production wheel. However, if parts have been added to the wheel after sequences were assigned then not all part records might display in the worksheet. Run the Generate part records command to automatically generate any of the missing part records. Records are generated for all production wheels displaying in the Wheels worksheet.



Note: Depending on the worksheet's data settings, different production wheels will display in the worksheet. For more about data settings, the [Maestro User Guide \(Java client\)](#).

By default the part sequences in the new records are set to 0. You must then assign a part sequence to each part. See ["Assigning part sequences" on page 528](#).

► Generate part records

1. In the **Production Planning Parameters** workbook, click the **Wheels** worksheet tab.
2. On the workbook toolbar, click **Run Command** , and then click **Generate sequence records**.
3. Click **Yes**.

Assigning part sequences

When you assign sequence numbers to parts, you can assign one sequence at a time or multiple sequences while viewing all parts in that wheel.

► Assign a sequence to a part in a production wheel

1. In the **Production Planning Parameters** workbook, click the **Wheels** worksheet tab.
2. Select the row for the wheel you want to view part sequences for.
3. Click the **Add** button  in the **Part - Sequences** column.
4. In the **Assign Part Sequence** form, click the **Part: = Current sequence =** field and select a part in the list.
The sequence currently assigned to the part displays next to the part name. For example, =0= indicates that the part has no sequence assigned to it.
5. Type a new number for the part in the **New sequence** field.
6. Click **OK**.



Note: Parts assigned smaller sequence numbers are processed first.

► Assign sequences to multiple parts in a production wheel

1. In the **Production Planning Parameters** workbook, click the **Wheels** worksheet tab.
2. Select the row for the wheel you want to view part sequences for.
3. Click the value in the **Part - Sequences** column to open the **Part Sequences** worksheet.
4. In the **Part - Sequence** column, type a number for the sequence.
5. Repeat step 4 for each part you want to assign a sequence to.
6. Save your changes.



Note: If you have two production wheels with the same name at different sites and you have accidentally selected the wheel at the wrong site, you can change the site for the wheel using the **Wheel - Site** column.

Flexible deferrals

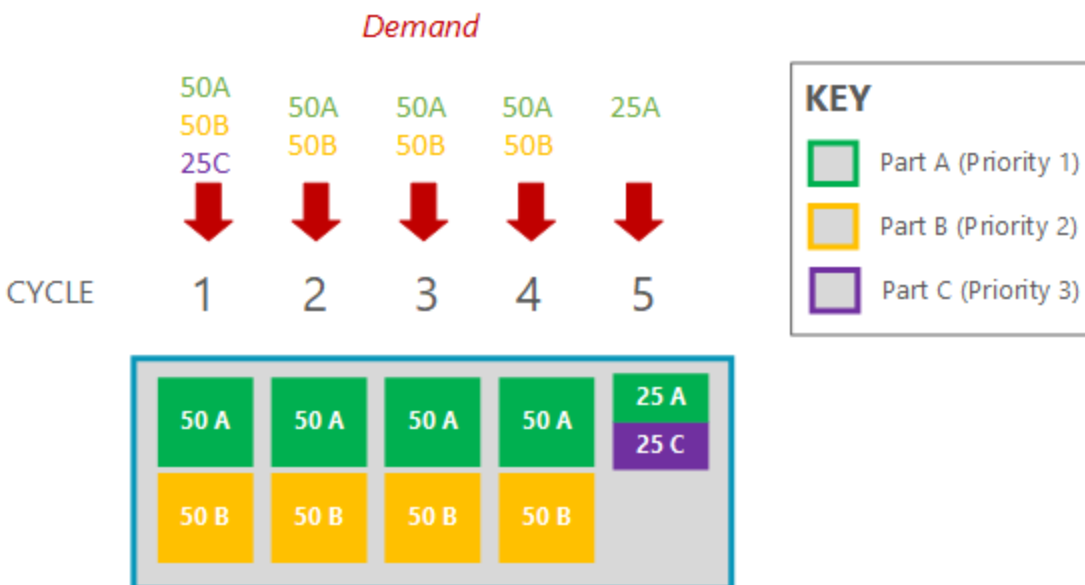
In situations where your production system is operating at, or close to, capacity, lower priority parts might become indefinitely deferred. This is because all capacity in the system is consumed by higher priority parts. For some of those deferred planned orders, you might want them to be prioritized in production without having to change part priorities that might not be feasible due to group changeover sizes.

A deferral rule on the part sources for these orders can be set to flexible to prioritize those orders even if they are low priority. The orders are shifted to the start of eligible cycles, after any non-reschedulable SRs in those cycles.

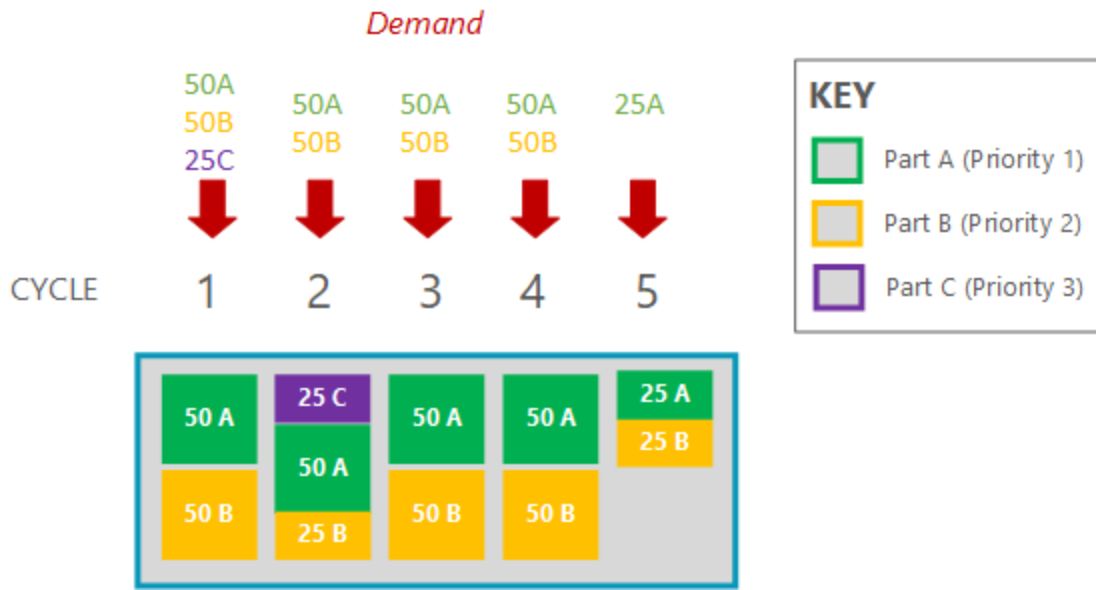


Note: Eligible cycles are cycles with available constraint that match the recurrence assigned to the part source.

In the diagram below, the demand for Part C is in cycle 1, but its production is deferred until cycle 5 because of capacity constraints. If demand for Parts A and B were to increase, Part C could become indefinitely deferred.



When Part C is set to flexible deferral, then it is shifted into cycle 2 where there is eligible constraint. Production of Parts A and B are shifted accordingly to account for 25 less constraint in cycle 2.

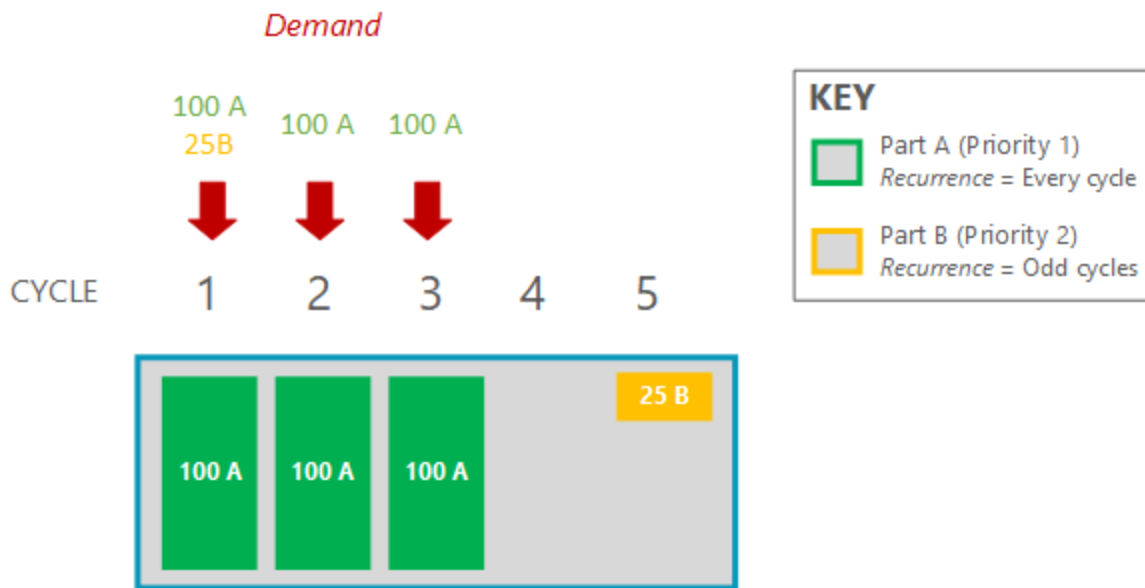


Note: Firm orders are not affected and retain their position in the cycle. Planned orders that are not set to flexible do not change behavior.

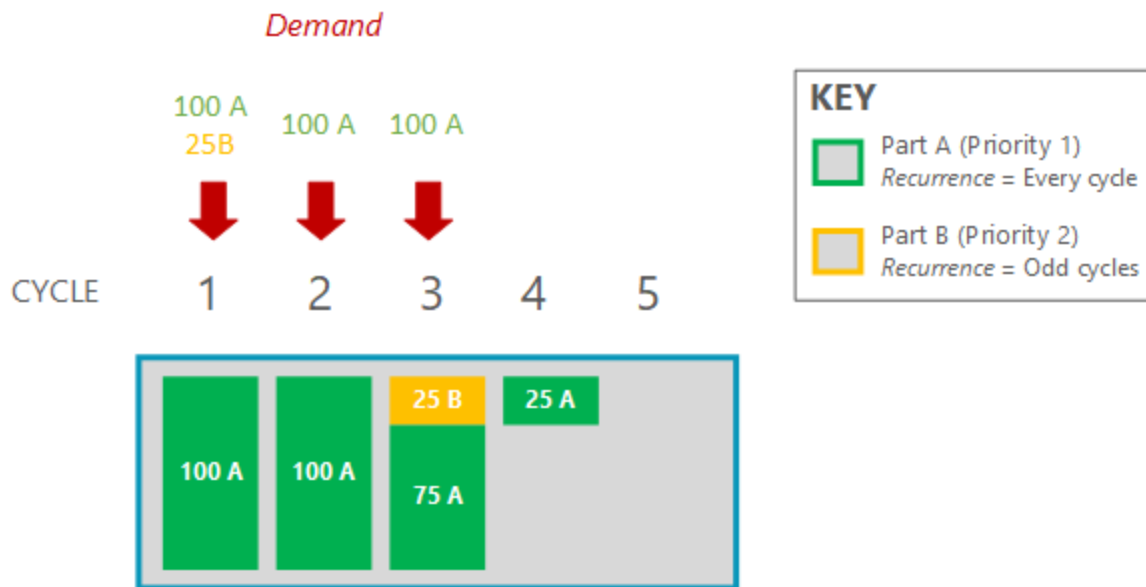
Recurrence and flexible deferrals

For orders set to flexible deferrals, the recurrence assigned to the part is still respected.

In the diagram below, demand for part A consumes all constraint in cycles 1 - 3. B cannot be produced until cycle 5 because of its assigned recurrence.



When B is set to flexible deferral, it is shifted to the first eligible cycle that matches its assigned recurrence: cycle 3.

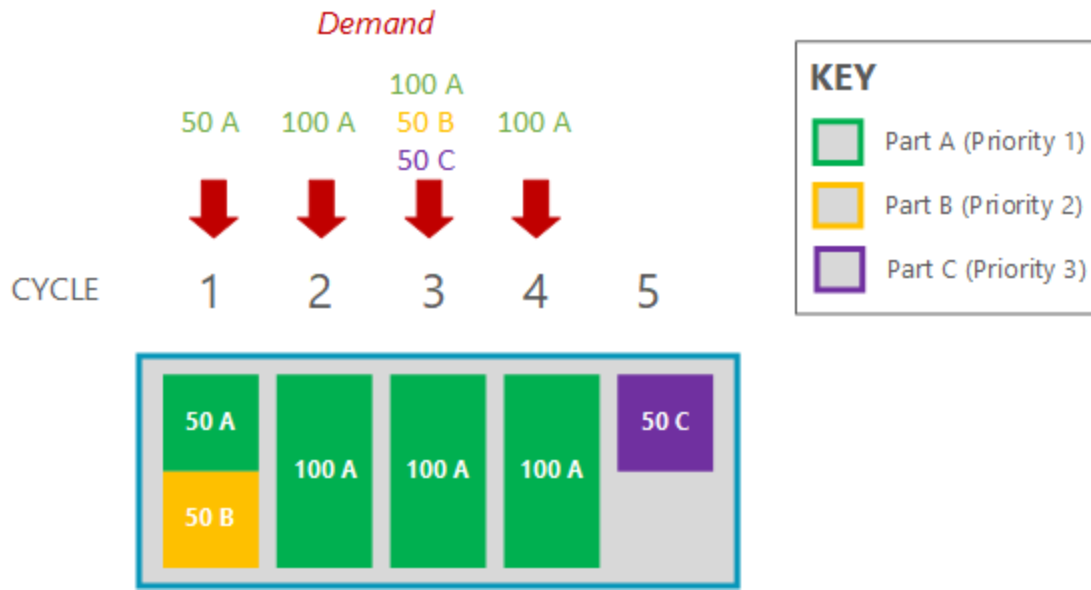


When you have multiple planned orders set to flexible deferral and the orders have different due dates, the most deferred order is prioritized. Otherwise, the orders are sorted by production priority. For more on how production is prioritized, see "[Production priorities](#)" on page 542.

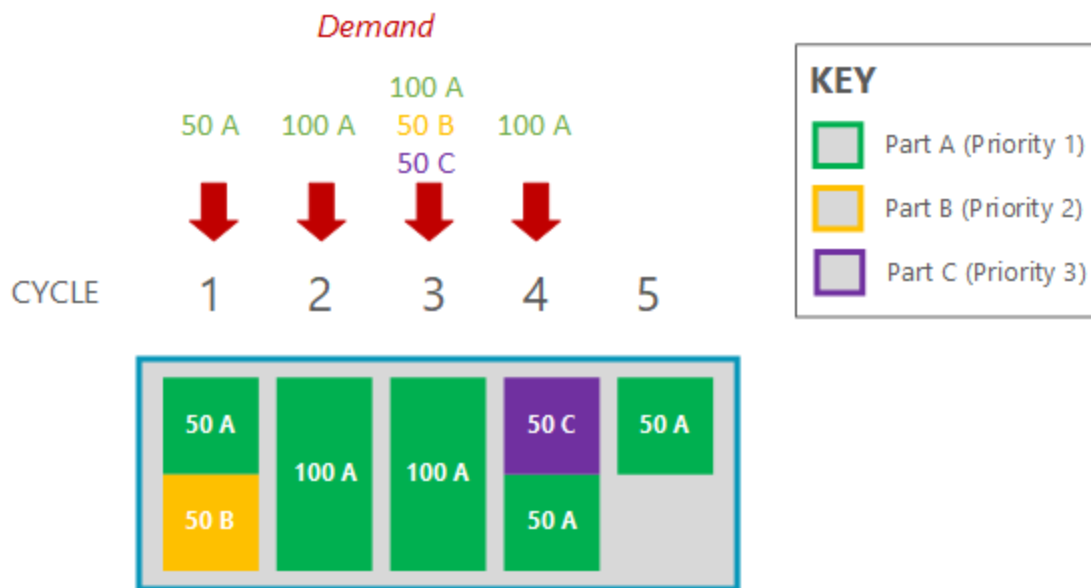
Pre-planning and flexible deferrals

You can set a part to flexible deferral and also assign it to be pre-planned. In this situation, Maestro pre-plans production and then any deferred planned orders set to flexible are shifted within the production wheel.

In the diagram below, pre-planning is first applied to the production wheel and because there is available capacity in cycle 1, part B is produced in full to meet its demand from cycle 3. Part C is a lower priority and is therefore pushed out to cycle 5 despite its demand being in cycle 3.



Next, because part C is set to flexible deferral, it is shifted to the earliest eligible cycle, cycle 4.



Note: Pre-planning is applied only once to the wheel, before any deferred orders are shifted.

► Set a part to use flexible deferrals

To set planned orders to use flexible deferral, the deferral rule is configured at the part source level. By default, all part sources are set to *Normal* where deferred planned orders are placed in the next eligible cycle based on part sequence and group priority. When you change the rule to

Flexible, deferred orders for that part source are placed after Non-Reschedulable SRs at the start of the next eligible cycle.

1. In the **Production Planning Parameters** workbook, click the **Source Configurations** worksheet tab.
2. For the part you want to make flexible, click in the **Deferral Rule** column and change the value to **Flexible**.

Production recurrence

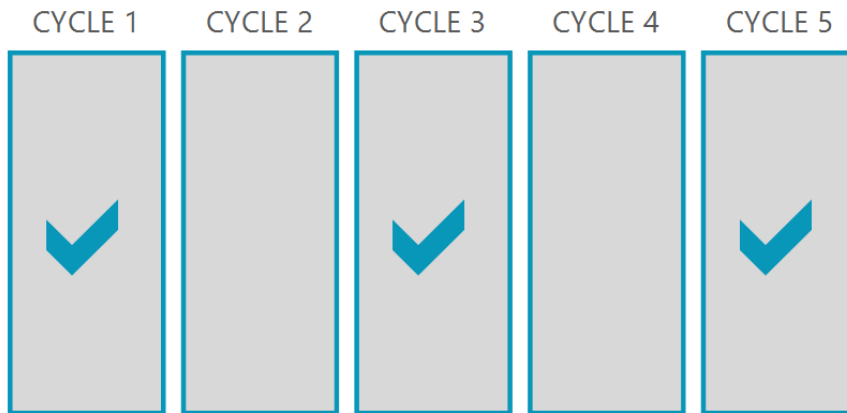
Production recurrences determine how often a part can be produced in a production wheel. The recurrence can be assigned to a specific part or to a segment associated with one or more parts.

Use the following steps to create or update a production recurrence:

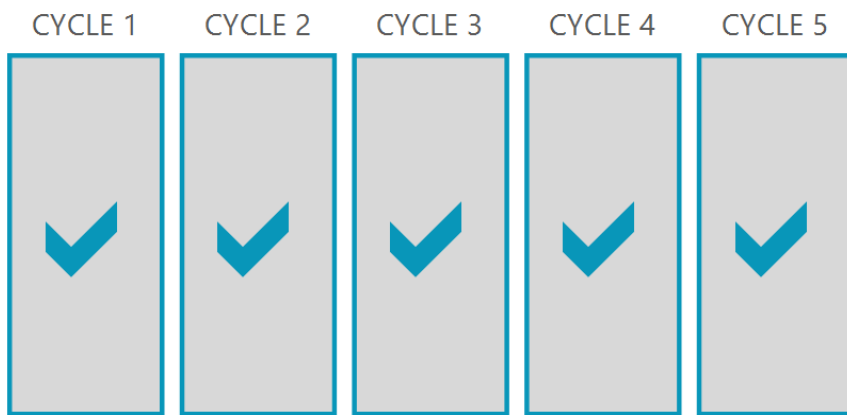
Step	Description
1. Define a production recurrence	Create a new recurrence or modify an existing recurrence. See "Defining production recurrences" on page 533 .
2a. If you use segmentation, assign the recurrence to a segment and then assign the part sources	If you are using segmentation strategies in your production, assign the production recurrence to a segment and then run the command to associate part sources to that production recurrence. See "Assign a production recurrence to a segment" on page 536 and "Associate part sources with a production recurrence" on page 537 .
2b. If you are not using segmentation, manually assign a recurrence to a part.	If you are not using segmentation strategies, directly assign a recurrence to a part. By default the part sources for the part are assigned to the recurrence. See "Assign a production recurrence to a part" on page 537 .

Defining production recurrences

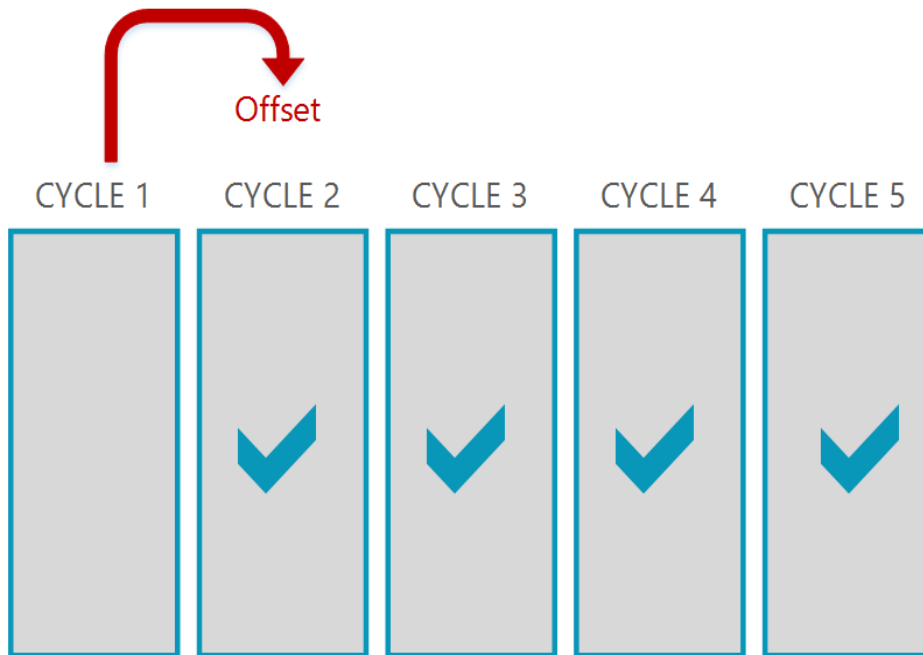
When you define a production recurrence, you determine which cycles in a production wheel the part can be produced in. For example, a part that is assigned a recurrence of 2 means that the part can be produced in every second cycle of a wheel.



A recurrence of 1 means that production can take place in every cycle of a wheel.



Optionally you can also specify an offset for the recurrence. The offset is the number of cycles the start of production for a part is delayed. For example, if Part A has a recurrence of 1 and an offset of 1 then that part can be produced in every cycle but it starts in the second cycle of the wheel.



Production recurrences and supply order types

All scheduled receipts are produced until the order is complete and can continue into subsequent cycles even if this breaks assigned production recurrence. As shown in the image below, to identify which production cycles have a break in recurrence, review the Load by Cycle Start Date worksheet in the Production Planning Manager workbook for constraint values shaded in gray.

Start x Production Planning Manager x

Non-reschedulable SRs use due date, not production recurrence, to determine which cycle to produce an order in. However, if the non-reschedulable SR has an assigned lead time, the

constraint date rule assigned to its part source determines if its production is initiated at its start date or its due date. If the cycle that the order is assigned to is out of recurrence, then the receipt is moved to the preceding cycle in the recurrence pattern.



Note: For large orders, non-reschedulable receipts can start in a cycle that fits the production recurrence but end in a cycle that breaks the recurrence pattern.

► Create or modify a production recurrence

1. In the **Production Planning Parameters** workbook, click the **Recurrences** worksheet tab.
2. To create a new recurrence, on the toolbar, click **Insert Record**
3. To modify a recurrence, select the row for the recurrence you want to modify.
4. In the **Insert Record - Production Recurrence** form, specify the following:
 - **Value:** The name for the production recurrence.
 - **Site:** The site for the production recurrence.
 - **Description:** A description of the production recurrence.
 - **Offset:** The number of cycles the production recurrence is offset by.
 - **Recurrence:** The cycles in the production wheel that can produce a part using this production recurrence.
5. Click **OK**.
6. Save your changes.

► Delete a production recurrence

For production recurrences that are no longer needed, you can remove the recurrence from your system.

1. In the **Production Planning Parameters** workbook, click the **Recurrences** worksheet tab.
2. Select the row for the production recurrence you want to remove.
3. On the toolbar, click **Delete Record**
4. Save your changes.

Assign a production recurrence

When you assign a production recurrence to a segment, all of the parts assigned to that segment are process in a production wheel using that recurrence. If you are not using segmentation strategies in your production, you can directly assign a production recurrence to a specific part.

► Assign a production recurrence to a segment

1. In the **Production Planning Parameters** workbook, click the **Segments** worksheet tab.
2. Select the row for the segment you want to assign a production recurrence to.
3. In the **Recurrence - Value** column, click the cell.

4. Select a recurrence in the list.
5. Optionally, change the site for the recurrence by clicking **Site** and selecting a different site in the list.
6. Save your changes.

► Assign a production recurrence to a part

1. In the **Production Planning Parameters** workbook, click the **Wheels** worksheet tab.
2. Select the production wheel the part is produced in.
3. Click the value in the **Part - Sources** column.
4. In the **Part Sources** worksheet, select the part you want to assign or change a recurrence for.
5. In the **Recurrence - Value** column, click the cell and select a new recurrence in the list.
6. Optionally, change the site for the recurrence by clicking **Site** and selecting a different site in the list.
7. Save your changes.

Associate part sources with a production recurrence

When you assign parts to a production recurrence, the parts sources for those parts are automatically associated with that recurrence.

1. In the **Production Planning Parameters** workbook, click the **Recurrences** worksheet tab.
2. Select the production recurrence you want to assign parts to.
3. To associate part sources from the same site as the selected recurrence, click the value in the **Part - Sources** column.
Only parts at that site display in the **Part Sources** worksheet.
4. To associate part sources from any site, click the **Add** button  in the **Part - Sources** column.
Parts for all sites display in the **Part Sources** worksheet.
5. For each part you want to assign to this recurrence, select the recurrence from the list in the **Recurrence - Name** column.
6. Save your changes.



Note: If you have two production recurrences with the same name at different sites and you have accidentally selected the recurrence at the wrong site, you can change the site for the recurrence using the **Production Recurrence - Site** column.

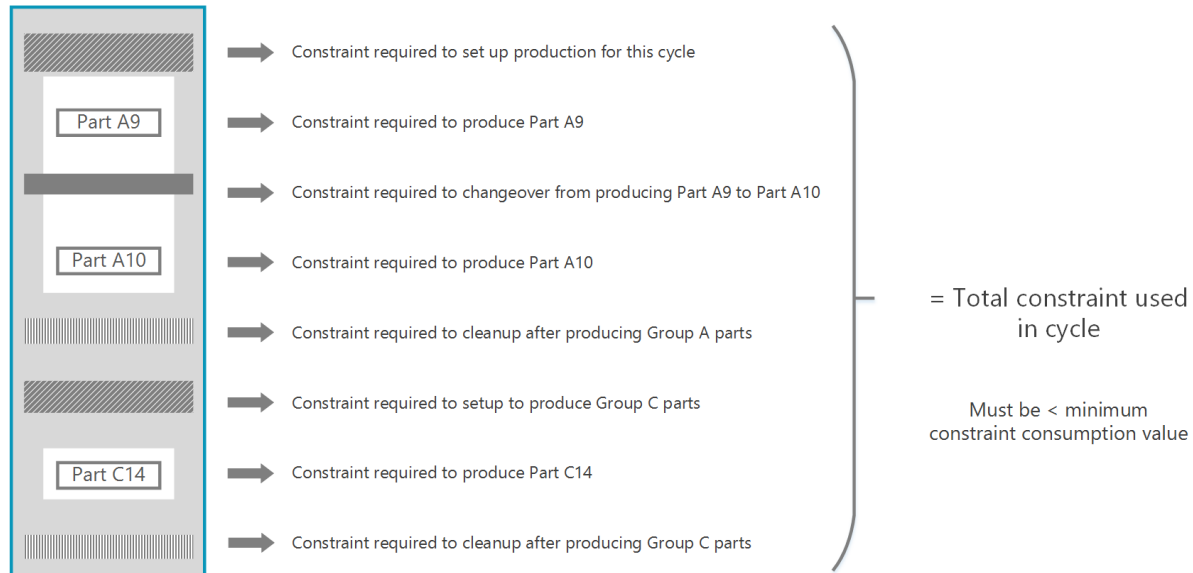


Note: Part sources using flexible deferral still follow assigned recurrence and deferred planned orders from those part sources are placed in the next recurring cycle with available constraint.

Assessing changeover

The Production Planning application provides you with the capability to plan the production of parts through a constraint in a predefined sequence. The total constraint available for production is divided among the cycles in each production wheel. Based on the parts produced in any given cycle, the amount of constraint required by a cycle varies.

The diagram below illustrates the different constraints that make up the total constraint required in the cycle.



Each cycle is limited by a minimum constraint consumption value. This is the minimum fraction of constraint that should be consumed in a cycle. When you are ignoring unused constraint in a production wheel, this limit is ignored. See ["Handle unused constraint amounts" on page 523](#).

Assign changeover for a production group

When you create or modify a production group, you define the constraint required to setup, cleanup, and changeover when producing parts in that group, including the maximum constraint consumption amount that the group is allocated. See ["Maximum constraint consumption" on page 544](#).

► Define the constraint for setup

1. In the **Production Planning Parameters** workbook, click the **Groups** worksheet.
2. Select the production group you want to define constraint for.
3. Type a value in the **Major - Setup** column.
4. Save your changes.

► Define the constraint for cleanup

1. In the **Production Planning Parameters** workbook, click the **Groups** worksheet.
2. Select the production group you want to define constraint for.
3. Type a value in the **Major - Cleanup** column.
4. Save your changes.



Note: You can override the amounts of constraint required for setup and cleanup for specific production groups. See ["Override changeover for a production group" on page 539](#).

► Define the constraint for changeovers

1. In the **Production Planning Parameters** workbook, click the **Groups** worksheet.
2. Select the production group you want to define constraint for.
3. Type a value in the **Minor Changeover** column.
4. Save your changes.

► Define the minimum constraint consumption quantity

1. In the **Production Planning Parameters** workbook, click the **Groups** worksheet.
2. Select the production group you want to define constraint for.
3. Type a value in the **Maximum Constraint Consumption** column.
4. Save your changes.

Override changeover for a production group

Constraint is used to model both minor and major changeovers in production wheels.

Major changeovers can be one of the following:

- **Setup:** the amount of constraint required to set up production for a group or cycle.
- **Cleanup:** the amount of constraint required to clean up production after a group or cycle has completed.

You can override any of the values specified for major changeovers by setting fixed setup and cleanup amounts between specified production groups.

For example, 4 different production groups are processed on a production wheel: Production Group A, B, C, and D. Group A, B, and D are produced on a regular basis and have slightly variable changeover amounts depending on the quantities being produced. Group C is only produced on special order and requires a thorough cleanup after production to minimize cross-contamination of materials. You can set the setup and cleanup amounts for Group C so that whenever it is produced in the wheel, those constraint amounts override any values that might have been input.

The override must be set for changing production from Group C to Groups A, B, and D and for changing production to Group C from Groups A, B, and D.

Production Wheels		Production Groups		Override Major Changeovers		Production Recu	
	Production Group			Cleanup	Setup		
	From	Site	To				
1	Group A	Bern	Group C	10	7		
2	Group B	Bern	Group C	10	5		
3	Group C	Bern	Group A	20	10		
4			Group B	25	15		
5			Group D	20	10		
6	Group D	Bern	Group C	12	10		

You can override any of the values specified for major changeovers by setting fixed setup and cleanup amounts between specified production groups.

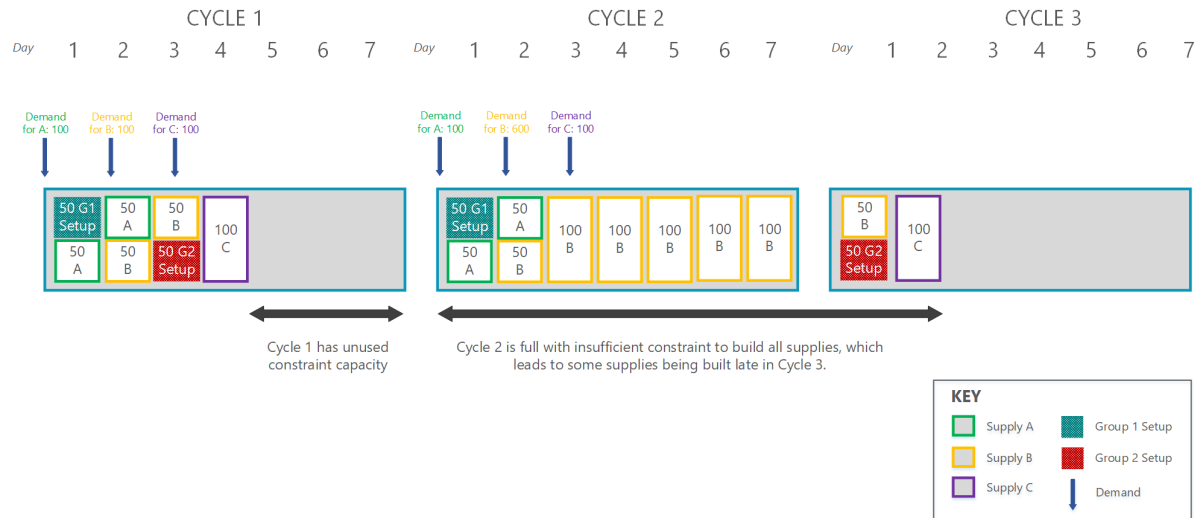
► Override changeover values for a production group

1. In the **Production Planning Parameters** workbook, click the **Override Major Changeovers** worksheet.
2. On the toolbar, click **Insert Record** .
3. In the **Insert Record - Override Major Changeover** form, select a site in the **Site** control.
4. For **From Production Group**, select the production group production is changing over from.
5. For **To Production Group**, select the production group production is changing over to.
6. For **Cleanup**, type a value for the amount of constraint required to clean up the production wheel after producing the production group.
7. For **Setup**, type a value for the amount of constraint required to set up production on the wheel for this production group.

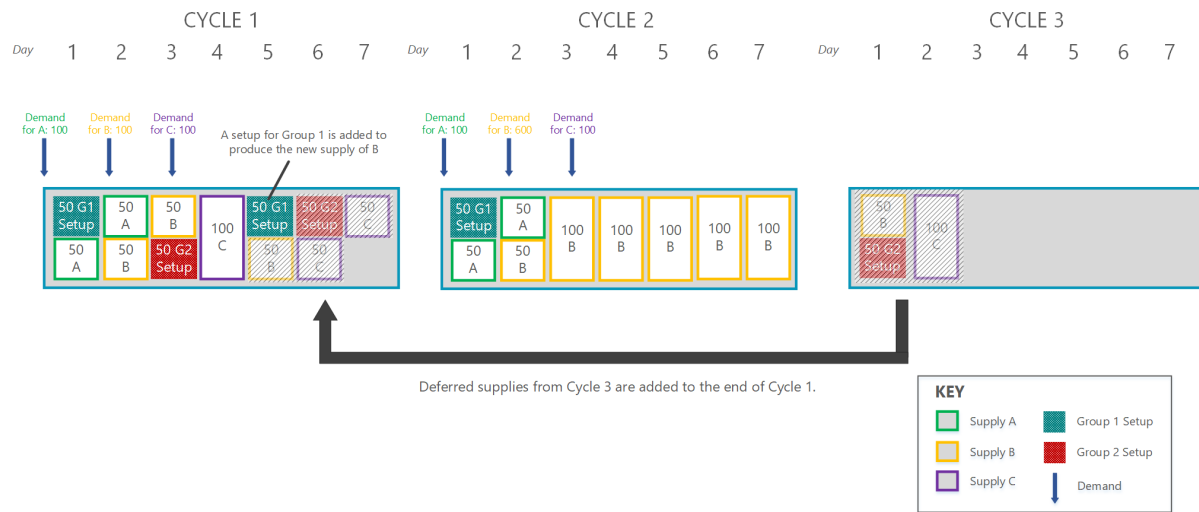
Pre-planning and production wheels

When pre-planning is enabled in your system, it improves capacity utilization and reduces the incidence of inventory stockouts. Pre-planning is triggered when insufficient constraint is detected in the current production cycle and supplies are preplanned and then resorted.

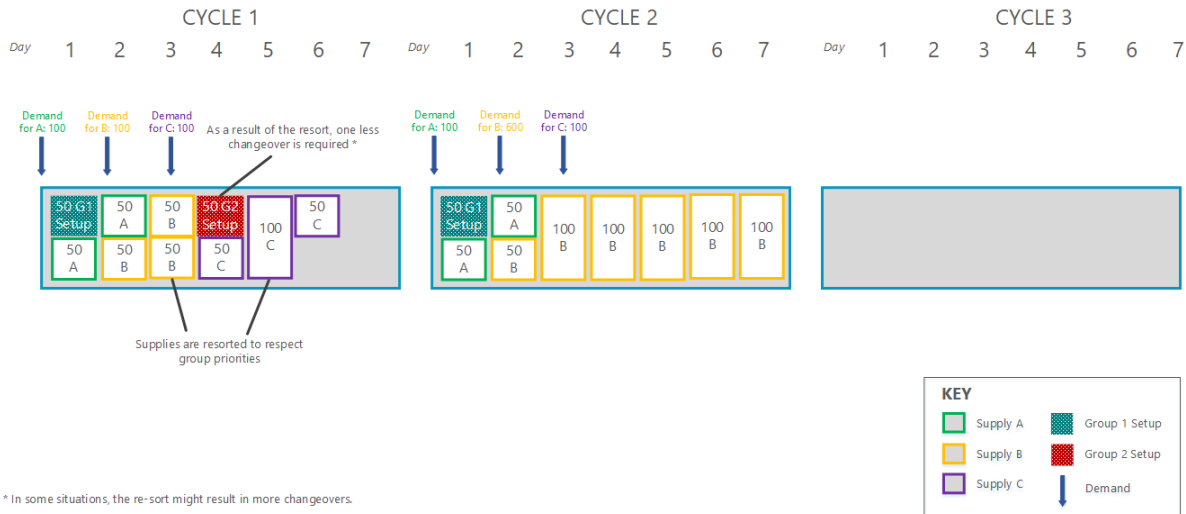
In the example below, a production wheel has cycles of 7 days and 100 units of constraint available each day. The diagram shows the setup of how part supplies are allocated in each cycle before pre-planning is applied. Cycle 1 has unused constraint capacity whereas Cycle 2 is full and must build some of its supplies late in Cycle 3.



The first stage of pre-planning moves deferred or ignored supplies with unused constraint to the end of previous cycles. The deferred supplies from Cycle 3 are moved to the end of Cycle 1 and a new setup (Group 1 Setup) is added to produce the first moved supply (Supply B).



The second stage of pre-planning is resorting the moved supplies by priority. The added supplies in Cycle 1 are resorted so that all Supply B and Supply C are produced together. As a result, the Group 1 Setup changeover added in the last stage is no longer required and is removed.



Pre-planning is enabled by configuring the **PrePlanLimit** and the **AbsolutePrePlanLimit** values on the PartSource table. See the [Maestro Data Model and Algorithm Guide \(Java client\)](#).

Pre-planning and supply order types

Scheduled SRs and non-reschedulable SRs cannot be preplanned. Instead, they consume constraint based on their start or due date and continue production until completed.

Pre-planning and flexible deferrals

Pre-planning is applied to a production wheel and then any deferred planned orders are shifted to eligible cycles with available constraint that match the recurrence assigned to the part source.

Production priorities

In addition to assigned production recurrences, you can set priorities for wheels, parts, and groups to further optimize when parts are produced:

- **Production Wheel Priority:** You can set the order that a production wheel is processed when multiple production wheels are scheduled to run in the same time period. When two or more wheels are assigned the same priority, Maestro uses the wheel name to determine which wheel to run first. Names are sorted alphabetically.
- **Part sequencing:** You can order a sequence of production by assigning sequence numbers to parts. Parts assigned a part sequence value of 1 are produced first, parts assigned a sequence of 2 are produced after all the sequence 1 parts are completed, and so forth. See ["Part sequencing" on page 526](#).
- **Production Group Priority:** You can set the order that a production group is processed when there are multiple production groups running on the same production wheel. The priority assigned to a group is overridden if sequencing is applied to any parts belonging to the group.

It is recommended that at minimum, you use part sequencing or production group priority to set priorities for all parts in a production wheel. This ensures that supplies remain grouped together and processed in the order intended.



Note: Different supply order types are prioritized differently in a production wheel. See ["Planning for different supply order types" on page 507](#).

► Set the priority for a production wheel

You can set the order that a production wheel is processed when multiple production wheels are scheduled to run in the same time period.

1. In the **Production Parameters** workbook, click the **Wheels** worksheet tab.
2. In the worksheet select the production wheel you want to modify.
3. Type a new value in the **Wheel - Priority** column.
4. Save your changes.

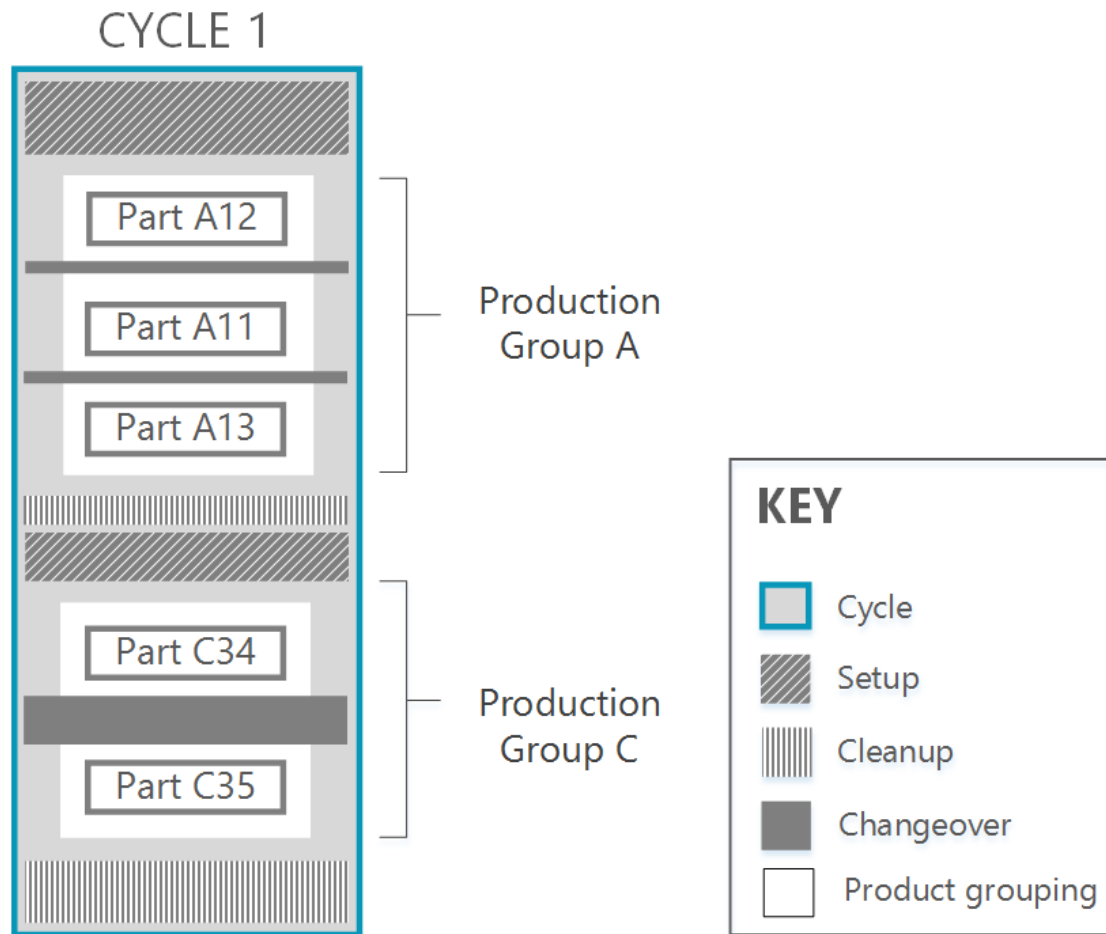
► Set the priority for a production group

You can set the order that a production group is processed when multiple groups run in the same production wheel.

1. In the **Production Parameters** workbook, click the **Groups** worksheet tab.
2. In the worksheet select the production group you want to modify.
3. Type a new value in the **Group - Priority** column.
4. Save your changes.

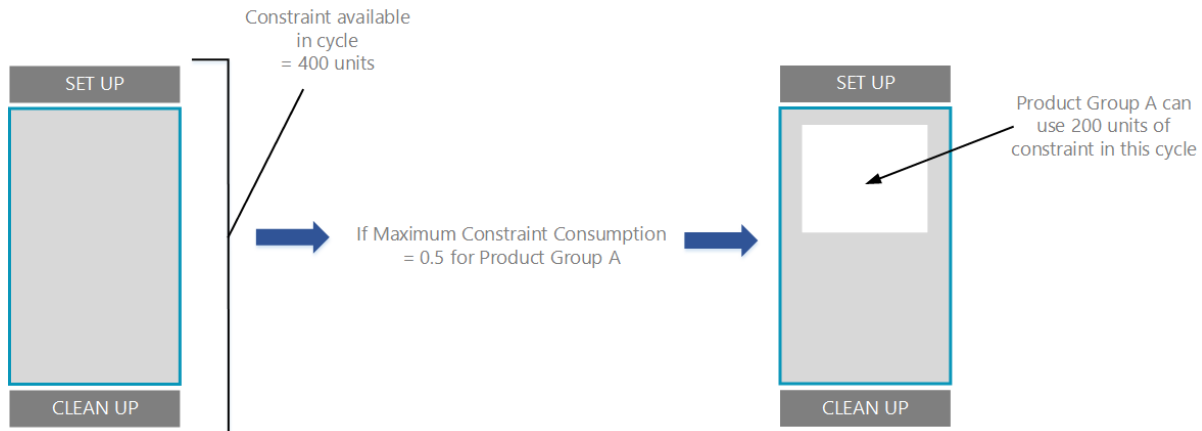
Production groups

To increase production efficiency and minimize changeover costs, you can organize parts into production groups. Production groups are associated with a specific site and have an assigned priority. See ["Production priorities" on page 542](#). Each group has a defined amounts of constraint allocated to the setup and cleanup for the group and for minor changeovers between parts within the group. See ["Assessing changeover" on page 538](#).



Maximum constraint consumption

Each group also has a maximum amount of constraint that can be consumed by the production group in a single production wheel. This value is expressed as a fraction of the constraint available in a cycle. For example, if you set the maximum constraint consumption to 0.5 and a cycle of the production wheel has 400 constraint available then the product group can only use 200 constraint. When this maximum limit is reached you cannot plan for parts in this product group in this cycle.



The constraint used in setup, changeovers, and cleanup is included in this maximum constraint consumption limit. This maximum value is reset at the start of each cycle in the production wheel.

Use the following steps to create or update a production group:

Step	Description
1. Create or modify a production group	Create a new production group or modify an existing production group. See "Defining production groups" on page 545 .
2. Associate part sources with the group	Parts assigned to be produced in a production group automatically associate part sources with that group. See "Associate part sources with a production group" on page 546 .

Defining production groups

When you create or modify a production group, you must define a variety of properties including the constraint required for each type of downtime in production.

For groups that are no longer needed, you can delete the group from your system.

► Create or modify a production group

1. In the **Production Planning Parameters** workbook, click the **Groups** worksheet tab.
2. To create a new group, on the toolbar, click **Insert Record** .
3. To modify a group, select the record for that group.
4. In the **Insert Record - Production Group** form, specify the following:
 - **Name:** The name of the production group. If you are typing a new name, you cannot use the same name as any other production group in your system.
 - **Site:** The site for the production group.
 - **Description:** A description of the production group.

- **Priority:** The number value for the priority. This priority determines what order the production group is processed when there are multiple production groups running in the same production wheel.

Changeovers

- **Major Setup:** The amount of constraint required to prepare the shared resource/line/machinery to produce the parts in this group.
- **Major Cleanup:** The amount of constraint required to clean the shared resource/line/machinery after the production of all parts in this group is completed.
- **Minor Change Over:** The amount of constraint required to shift between producing one part in this group to another part in the group.

Consumptions

- **Maximum Constraint Consumption:** The maximum fraction of constraint available for this group in a cycle. See "[Maximum constraint consumption](#)" on page 544.

5. Click **OK**.
6. Save your changes.

► Delete a production group

If a production group is no longer required or updating it is not feasible, then you can delete the production group record from your Maestro system.

1. In the **Production Planning Parameters** workbook, click the **Groups** worksheet tab
2. Select the row for the production group you want to delete.
3. On the toolbar, click **Delete Record** .
4. Save your changes.

Associate part sources with a production group

Part sources are automatically associated with a production group when you assign parts to be produced in that group.

1. In the **Production Planning Parameters** workbook, click the **Groups** worksheet tab.
2. Select the production group you want to associate parts with.
3. To associate part sources from the same site as the selected group, click the value in the **Part - Sources** column.

Only parts at that site display in the **Part Sources** worksheet.

4. To associate part sources from any site, click the **Add** button  in the **Part - Sources** column.

Parts for all sites display in the **Part Sources** worksheet.

5. For each part you want to produce in this group, select the group from the list in the **Group - Name** column.
6. Save your changes.



Note: If you have two production groups with the same name at different sites and you have accidentally selected the group at the wrong site, you can change the site for the group using the **Production Group - Site** column.

Review constraint usage

Constraints are fixed or variable factors that represent the capacity of a process or supplier. For example, a constraint might be the number of units a production line can produce daily. For more about constraints, see the [Maestro Data Model and Algorithm Guide \(Java client\)](#).

Each cycle in a production wheel is allocated constraint based on which parts are produced in the cycle. For more on how constraint is divided in a cycle, see ["Assessing changeover" on page 538](#).

Use the following steps to review constraint usage in production wheels:

Step	Description
Explore how constraints are allocated in production wheels in your system	Review constraint data by load, cycle number, or cycle start date or focus on a specific production wheel or the production setup of a wheel. See "Viewing constraint data" on page 547 .
Compare how constraint load is used over time or by cycle	Identify how constraint is used in the cycles of a production wheel. See "Compare load utilization over time or cycle count" on page 550 .
Communicate about production	Communicate about specific production wheels, parts, or constraints in a collaboration, message, or by sharing a scenario. See "Communicate about production" on page 552 .
Identify cycles where constraint capacity has been exceeded	Identify which cycles have consumed more than the allocated constraint available for the production wheel. See "Identify cycles with excessive constraint" on page 551 .

Viewing constraint data

The Production Planning Manager workbook provides you with an overview of the constraints used by parts produced in production wheels. You can change how constraint data displays by changing how the data is grouped, focus on a single production wheel, or see details on how production is set up. At any time you can reset the workbook layout to its original view.

► Focus on a production wheel

View constraint details for all production wheels in your system or focus on a single production wheel.

1. In the **Production Planning Manager** workbook, click **Wheel:** on workbook toolbar.
2. Select = **All** = or a wheel name in the list.

► See more about a constraint

You can drill to more detailed information about the constraint assigned to a production wheel.

1. In the **Production Planning Manager** workbook, click **Details** on the workbook toolbar.
2. Do one of the following:
 - Click a value in the **Wheel - Constraint** column.
 - In the **Load Detail** worksheet, click a value in the **Constraint - Name** column.
3. Select one of the following:
 - **Constraint Properties:** See detailed data about constraints and how they are configured.
 - **Constraint Manager:** See a summary of constraints and the scheduled usage for those constraints. You can drill to different levels of detail about constraints from this workbook.

► Show production details

To better understand how each part is configured in a production wheel, you can see more about the production wheel, group, and recurrence associated with each part.

- In the **Production Planning Manager** workbook click **Details** on the workbook toolbar.

► Identify the type of supply order

To better understand how a production wheel is configured, you can identify whether the supply order for a part is a planned order or a scheduled receipt.

1. In the **Production Planning Manager** workbook, click a **Total** value to open the **Load Detail** worksheet.
2. Click **Details** on the worksheet toolbar.
3. Scroll across to the **Supply Order - Type** column.



Note: Specific types of SRs are not identified, only whether an order is a planned order or a scheduled receipt.

► Reset the workbook layout

As you are exploring the data in the Production Planning Manager workbook, you might drill to multiple worksheets. You can close all the extra worksheets and return to focus on the main worksheet in the workbook by resetting the layout.

- In the **Production Planning Manager** workbook click **Reset** on the worksheet toolbar.

Constraint grouping

Constraint quantities in the worksheet can be grouped in one following ways:

- **By load date:** constraint is grouped into date buckets. The buckets are configured in the Bucket Settings. For more information, see ["Define date buckets for load data" on page 549](#).
- **By cycle number:** constraint is grouped by cycle. The cycle number counts the number of cycles the wheel has run from its effective in date.
- **By cycle start date:** constraint is grouped into date buckets based on cycle start dates.

► Change how constraint quantities are grouped

1. In the **Production Planning Manager** workbook, click **View by** on worksheet toolbar.
2. Based on how you want to see constraint group, select **Load Date**, **Cycle Number**, or **Cycle Start Date**.

You can identify how constraint loads are utilized by drilling to a constraint chart. Depending on how the constraint is grouped in the worksheet, different charts display. For more information, see ["Compare load utilization over time or cycle count" on page 550](#).

► Define date buckets for load data

When you view constraints grouped by load date, you can define which date buckets the data displays for. The workbook author sets the anchor date and you select how many buckets to display before and after this anchor date.

1. In the **Production Planning Manager** workbook, click **View by: Cycle Number** on the workbook toolbar.
2. On the workbook toolbar, click **Bucket Settings** .
3. To include a past bucket, select the **Include Past bucket** checkbox.
4. In the **Before anchor date** area, select the number and size of buckets you want to display before the anchor date.
5. In the **After anchor date** area, select the number and size of buckets you want to display after the anchor date.

The first bucket specified after the anchor date will contain the anchor date. For example, if the first bucket is a weekly bucket it will include data from the week that includes the anchor date.
6. To include a future bucket, select the **Include Future bucket** checkbox.
7. To insert buckets when bucket sizes change, select the **Automatically add buckets when transitioning between bucket sizes** checkbox.
8. Click **OK**.



Note: If you select past or future for any of the buckets then all of the dates earlier or later than the specified dates display in a single past or future bucket.

► See more about a constraint load

You can see more information about the constraint load in a bucket such as the sum and total percent utilization of the load in the production wheel. If production details are already displayed, the Load Detail worksheet displays more information about the production wheel, cycle, constraint, production group, part, and supply order for the load. For more information, see ["Show production details" on page 548](#).

1. In the **Production Planning Manager** workbook, click **View by: Load Date** on the workbook toolbar.
2. Click a **Total** value for any load bucket.

► View the impact of the constraint plan on inventory

When viewing constraint data grouped by load dates, you can drill to details on demand, supply, and inventory data for a part and review how the constraint plan affects inventory quantities in relation to both demand and supply.

1. In the **Production Planning Manager** workbook click **View by: Load Date** on the workbook toolbar.
2. Click a value in the **Part** column.

Compare load utilization over time or cycle count

You can identify how constraint is used in a production wheel by drilling to a chart from the wheel name. Depending on how the constraint is grouped, different charts display.

► View constraint load by date

Analyze how constraint load is used over a period of time in the Constraint Load by Date worksheet. Productive periods of the production wheel display in teal. Unproductive periods of the production wheel display in gray.

- **X-axis:** date buckets
 - **Y-axis:** summarized constraint quantities, measured in constraint units
1. In the **Production Planning Manager** workbook, set the **View by** control to **Load Date**.
 2. Click the **Name** for a wheel.
 3. Select **Constraint Load by Date** in the list.
 4. Hover over a bar to view the date bucket, type, and total value for that load.

► View constraint utilization by cycle number

Analyze constraint utilization by cycle in the Constraint Utilization by Part (Cycle) chart. The summarized constraint is divided into the segments associated with each part. The parts are colored the same as their assigned segment: green, yellow, blue, or red. Parts not assigned to a segment display in teal. Unproductive periods of the production wheel display in gray.

- **X-axis:** cycle numbers
 - **Y-axis:** the amount of constraint allocated to a cycle, measured in constraint units
1. In the **Production Planning Manager** workbook, set the **View by** control to **Cycle Number**.
 2. Click the **Name** for a wheel.
 3. Select **Constraint Utilization by Part (Cycle)** in the list.
 4. Hover over a section of a bar to view the cycle number, part label, and total utilization for that part.

► View constraint utilization by cycle start date

Analyze constraint utilization by the start date of each cycle in the Constraint Utilization by Part (Cycle Start Date) chart. The summarized constraint is divided into the segments associated with each part. The parts are colored the same as their assigned segment: green, yellow, blue, or red. Parts not assigned to a segment display in teal. Unproductive periods of the production wheel display in gray.

- **X-axis:** cycle start dates
 - **Y-axis:** the amount of constraint allocated to a cycle, measured in constraint units
1. In the **Production Planning Manager** workbook, set the **View by** control to **Cycle Start Date**.
 2. Click the **Name** for a wheel.
 3. Select **Constraint Utilization by Part (Cycle Start Date)** in the list.
 4. Hover over a section of a bar to view the cycle start date, part label, and total utilization for that part.



Note: At any time, you can return the workbook to its original layout and close extra worksheet tabs. For more information, see ["Reset the workbook layout" on page 548](#).

Identify cycles with excessive constraint

You can identify which cycles in a production wheel have consumed more than the allocated constraint available in the wheel. This can happen if the cycle includes large scheduled SRs orders. Once production of a scheduled SR begins, it continues until the order is complete,

irrespective of constraint limitations. Constraint utilization sums for a wheel that exceed 100% are wheels with cycles that consume more than the allocated constraint.

1. In the **Production Planning Manager** workbook, click a **Total** value for a **Load Date**, **Cycle Number**, or **Cycle Start Date** to open the **Load Detail** worksheet.
2. In the **Utilization** column, a **Sum** row value that exceeds 100% identifies that the cycle is consuming more constraint than is available.

Communicate about production

Communicate with the person responsible for a production wheel, constraint, or part by:

- Starting a collaboration
- Sending a message
- Sharing a scenario with them

► Start a collaboration about a production wheel

1. In the **Production Planning Manager** workbook, click the contact card  in the **Wheel - Responsible** column.
2. In the contact card, click **Collaborate**.
3. Do one of the following in the **Share** dialog box:
 - Click **New Collaboration**.
 - Click a collaboration in the **Existing Collaborations** area.
4. If you want to discuss the constraint or part in an existing collaboration, click the name of an existing collaboration.

A link to the **Production Planning Manager** workbook and a screenshot of the workbook are automatically included in the collaboration post.

5. Optionally, type explanatory or additional text in the post.
6. Click **Post**.

► Start a collaboration about a constraint or a part

1. In the **Load Detail** worksheet, click the contact card  in one of the following columns:
 - **Constraint - Responsible**
 - **Part - Responsible**
2. In the contact card, click **Collaborate**.
3. Do one of the following in the **Share** dialog box:
 - Click **New Collaboration**.
 - Click a collaboration in the **Existing Collaborations** area.
4. If you want to discuss the constraint or part in an existing collaboration, click the name of an existing collaboration.

A link to the **Production Planning Manager** workbook and a screenshot of the workbook are automatically included in the collaboration post.

5. Optionally, type explanatory or additional text in the post.
6. Click **Post**.

► **Send a message about a production wheel**

1. In the **Production Planning Manager** workbook, click the contact card  in the **Wheel - Responsible** column.
2. In the contact card, click **New Message**.
3. Optionally, select additional recipients of the message.
4. Add text in the **Message** box to explain why you are sending the message.
5. Optionally, add one or more links to the message or attach a report.
6. Click **Send**.

► **Send a message about a constraint or a part**

1. In the **Load Detail** worksheet, click the contact card in one of the following columns:
 - **Constraint - Responsible**
 - **Part - Responsible**
2. In the contact card, click **New Message**.
3. Optionally, select additional recipients of the message.
4. Add text in the **Message** box to explain why you are sending the message.
5. Optionally, add one or more links to the message or attach a report.
6. Click **Send**.

► **Share a scenario about a production wheel**

1. In the **Production Planning Manager** workbook, click the contact card  in the **Wheel - Responsible** column.
2. In the contact card, click **Share Scenario**.
3. In the **Share Scenario** box, click **Select Names** on the **Share** tab.
4. To add a user, on the Users select the person you want to add and then click one of the following:
 - **Modify**: people can change the data in the scenario by committing changes from other scenarios back into your scenario, or by editing the scenario directly if they have permission to edit shared scenarios.
 - **View only**: people can view data in the scenario but they cannot change its data.
5. To add a group, click the **Groups** tab and select the group you want to add.
6. Click **Modify** or **View only**. The type of permission you select is applied to all members of the group.
7. Repeat steps 4 or 5 for each user or group you want to share the scenario with.

8. Click **OK**.
9. If you want to send a message to people you are sharing the scenario with, in the **Share Scenario** dialog box click the **Notify** tab.
10. On the **Notify** tab, select the **Notify when given access to this scenario** checkbox, and then do one of the following:
 - In the **Subject** box type the subject for the message sent to people given access to the scenario.
 - In the **Message** box type the message sent to people given access to the scenario.
11. To add a link to the message, click the location where you want to insert the link and click **Insert Link** .
12. Optionally, in the **Notify me when** area, select one of the following checkboxes:
 - **Anyone responds**
 - **Everyone has responded**
 - **Anyone modifies data in this scenario**
13. Click **OK**.

► **Share a scenario about a constraint or a part**

1. In the **Load Detail** worksheet, click the contact card in one of the following columns:
 - **Constraint - Responsible**
 - **Part - Responsible**
2. In the contact card, click **Share Scenario**.
3. In the **Share Scenario** box, click **Select Names** on the **Share** tab.
4. To add a user, on the Users select the person you want to add and then click one of the following:
 - **Modify**: people can change the data in the scenario by committing changes from other scenarios back into your scenario, or by editing the scenario directly if they have permission to edit shared scenarios.
 - **View only**: people can view data in the scenario but they cannot change its data.
5. To add a group, click the **Groups** tab and select the group you want to add.
6. Click **Modify** or **View only**. The type of permission you select is applied to all members of the group.
7. Repeat steps 4 or 5 for each user or group you want to share the scenario with.
8. Click **OK**.
9. If you want to send a message to people you are sharing the scenario with, click the **Notify** tab.
10. On the **Notify** tab, select the **Notify when given access to this scenario** checkbox, and then do one of the following:

- In the **Subject** box type the subject for the message sent to people given access to the scenario.
 - In the **Message** box type the message sent to people given access to the scenario.
11. To add a link to the message, click the location where you want to insert the link and click **Insert Link** .
 12. Optionally, in the **Notify me when** area, select one of the following checkboxes:
 - **Anyone responds**
 - **Everyone has responded**
 - **Anyone modifies data in this scenario**
 13. Click **OK**.

Production Planning resources

The Production Planning application contains the following resources:

Resource	Use this resource to:
Production Planning Manager workbook	• Identify possible planning issues. • Review data about constraint usage for each production wheel you have configured. • Identify which parts are produced in a specific production wheel. • Identify the amounts of constraint applied to setup, clean up, change over, and variable processes in each production wheel. • View constraint usage by total load and by cycles. • Compare load utilization with actual supply. • Identify cycles using excessive constraint. • Identify cycles with breaks in production recurrence.
Production Planning Parameters workbook	• View and edit production wheel priorities and assigned constraints. • View and edit production group properties. • View and edit the recurrences assigned to segments. • View and edit the segmentation assigned to parts. • Set part sources to use flexible deferral for deferred planned orders.
Production Planning Segmentation workbook	• Analyze and group parts into segments. • Review demand data for segmented parts. • Change and override segment thresholds.